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REVIEW

A Comprehensive Survey on Fog Computing: Architectures, Techniques, Challenges, and Future Directions

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Abstract

The increasing scale and heterogeneity of Internet of Things (IoT) deployments have intensified the limitations of centralized cloud computing (CC), particularly in applications requiring low latency, efficient bandwidth utilization, and localized decision-making. Fog Computing (FC) has emerged as a distributed computing paradigm that places computational and storage capabilities closer to data-generating sources, thereby bridging the gap between edge devices and remote cloud infrastructures. This survey presents a systematic and critical review of FC research spanning the period from 2015 to 2025. Using a Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) inspired review methodology, the paper synthesizes existing studies to construct a structured taxonomy encompassing architectural models, service abstractions, task scheduling strategies, resource allocation mechanisms, Artificial Intelligence (AI)-enabled optimisation, security solutions, and energy-aware designs. Empirical results reported in the literature indicate that fog-based frameworks demonstrate reported improvements in prior studies cloud-centric systems, achieving latency reductions of up to 70%, bandwidth savings approaching 80%, and energy-efficiency improvements ranging from 15% to 50% across diverse domains including smart healthcare, vehicular networks, smart cities, and industrial IoT. Particular attention is given to learning-driven approaches, including Reinforcement Learning (RL) and Federated Learning (FL), which enable adaptive, scalable, and privacy-preserving fog operations. Based on identified research gaps, we propose FogAI-Optimizer, an integrated conceptual framework synthesized from systematic literature evidence that integrates multiple FC architectural and optimisation dimensions, green AI optimisation, edge intelligence, cloud integration, and dataset-driven validation into a cohesive model for next-generation fog systems. The survey concludes by identifying persistent challenges related to scalability, sustainability, and security, and outlines concrete future research directions, including FogAI-Optimizer implementation, Artificial Intelligence of Things (AIoT) integration, and 6 G-enabled fog deployments.

1 Introduction

In recent years, with the growth of the IoT to reach millions of interconnected devices that produce large amounts of data. Researchers and industry have come to the acknowledgement of the insufficiency of relying only on remote cloud infrastructure to do so in real-time. Such a requirement of instant response, autonomy and context-aware computing has resulted in the development of FC that reduces the centralized data centers to the network edge. FC has also grown out of pioneering efforts at Cisco and other industry giants to become a critical layer that enables latency-sensitive services in smart cities, industrial automation, vehicular networks, and immersive applications [1, 2]. As they are becoming more mature, FC is now being perceived as not just a complement to CC but an architectural paradigm shift, changing the nature of data management and response in the new digital ecosystems by decentralizing storage, analytics, and control over geographically distributed nodes.

1.1 Definition and Concept of Fog Computing

FC introduces a significant shift in managing the massive amounts of data generated by interconnected devices in

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contemporary computing environments [1]. FC differs from traditional distributed computing by bringing computational services closer to the network edge, as well as to the devices generating data. It directly interacts with data-generating sources such as IoT devices. Cisco defined FC in 2012 as an extension of CC that introduces intermediate layers where compute, storage, and networking resources are positioned between end devices and cloud data centers [3, 4]. This results in a hierarchical architecture with IoT devices at the bottom, fog nodes (routers, gateways, or local servers) in the middle, and the cloud at the top for large-scale storage and analysis [5].

FC enhances Quality of Service (QoS) parameters, like latency, bandwidth, and security via distributed processing with a geographical distribution across the network [6, 7]. It supports applications that are sensitive to geographical location, support mobility, and need ultra-low latency, making it suitable for applications on the Internet of Everything (IoE), where not only devices are connected to each other but people, processes, and data too [8]. Prominent characteristics include heterogeneity, scalability, and virtualisation [9].

1.2 Motivation for FC

The proliferation of IoT devices, with active connections estimated at over 14 billion as of 2024 and projected to exceed 25 billion by 2030, generates data volumes and latency requirements that cloud-centric architectures cannot sustain [10]. FC addresses these requirements by enabling computation at the network edge, reducing latency from 100 to 500 ms (cloud) to sub-10 ms (fog). FC addresses these shortcomings by providing a distributed computing solution that integrates the benefits of edge computing, providing high-speed computing capacity close to the edge [6].

1.3 Comparison with Cloud Computing

Conventional CC is based on centralised data processing, where end devices are dependent on remote servers, with computation and storage occurring on remote cloud servers. Although scalable and useful in several applications, this architecture introduces significant latency, bandwidth overhead, and reduced reliability in situations where a real-time or near-real-time response is required [11]. The extensive network paths between devices and cloud servers create communication bottlenecks, particularly in large-scale IoT systems. FC addresses these bottlenecks by moving computation closer to the data source using a distributed set of nodes, known as fog nodes, thereby decreasing latency, enhancing responsiveness, and mitigating network congestion [12, 13].

To avoid ambiguity throughout this work, high latency (100–500 ms) refers exclusively to cloud-centric architectures where data must traverse long network paths to remote data centres. FC, by contrast, achieves low latency (sub-10 ms) by processing data locally at fog nodes. Any reference to latency reduction percentages (50% to 70%) in this work indicates the improvement achieved by FC over CC baselines, not a limitation of FC itself.

Figure 1 displays the differences in the structure and performance of cloud-centric and fog-centric architectures. The IoT devices, such as cameras and wearables, also send their data in the cloud-centric model, transferring it to a centralized cloud server over long network paths. This strategy results in higher latency, increased bandwidth usage, scaling problems and possibly privacy concerns. Data is processed far from its source. Unlike that, the fog-centric architecture spreads the computation among the local fog nodes and middle-level network equipment. Here how FC addresses the shortcomings of traditional CC by employing a proximity-based and distributed processing paradigm is shown.

1.4 Applications and Use Cases

FC has applications that are geographically sensitive and require mobility support, as well as low latency, making it a promising design on the Internet of Everything (IoE), where devices, people, processes, and data are all connected [8]. FC has been implemented in different fields. In healthcare, for example, remote patient monitoring, ensuring that low-latency processing is necessary to conduct timely analyses and responses [11, 14]. In smart cities, FC enhances traffic control by utilising local data processing to streamline real-time data traffic flow [8]. FC is used to reduce predictive maintenance in factories, particularly for Industry 4.0 equipment, thereby minimising downtime [9]. Fog nodes are utilised in the local processing of sensors in the field of autonomous cars and video data to support real-time decision-making and collision avoidance [15].

1.5 Fog Computing Imitations

FC provides viable solutions to conventional centralized cloud architectures, but some of the challenges persist. To understand FC's significance, we must first acknowledge the shortcomings of cloud-centric systems. Cloud infrastructures are centralized in nature, and this normally causes huge delays in communication, especially when data needs to cover a long distance across geographically dispersed IoT sensors [16]. Moreover, due to the sheer amount of data generated by IoT, bandwidth may become congested, making continuous data transmission both energy-intensive and costly [7]. This becomes impractical for latency-sensitive

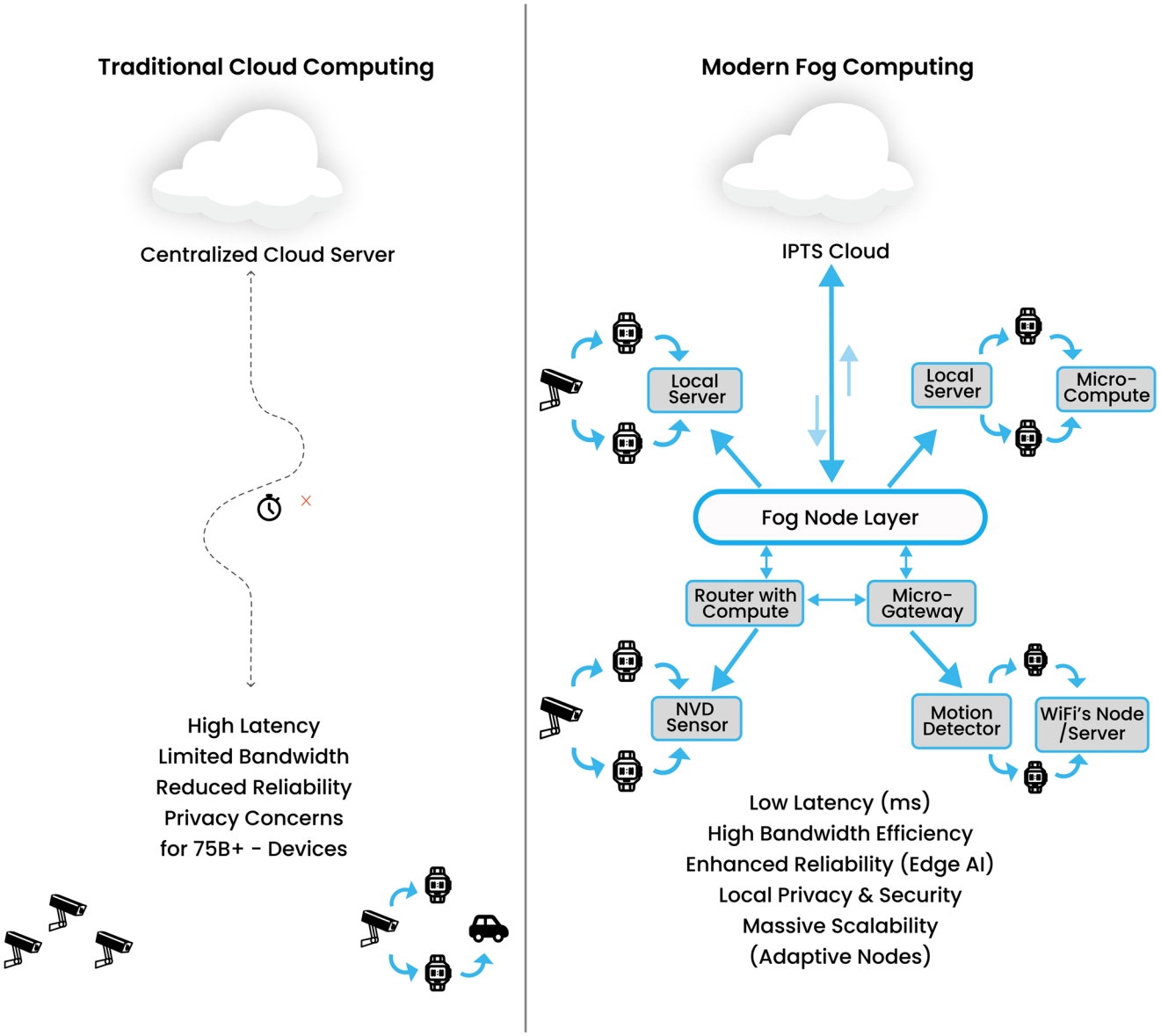


Fig. 1 Comparison of data flow and performance in cloud-centric vs. fog-centric architectures

applications that are latency-sensitive, such as real-time healthcare monitoring, which rely solely on cloud services [17]. Moreover, the transfer of sensitive data over open networks poses significant privacy and security threats, and research indicates that it remains challenging to scale cloud systems to support large-scale IoT implementations [6].

Fog nodes are often run on limited resources; a sensor-based adapter node can be in a worse position than a cloud server with respect to its computational capabilities [18, 19]. The fog environments are also characterized by a high level of hardware heterogeneity that raises complexity in the context of scheduling, resource allocation, and system management [15]. To illustrate, the authors of [8] examined the issue of resource visualisation and management in heterogeneous fog environments, constrained by energy. They found challenges in tracking, storing and maintaining long-term data devices access, specifically, healthcare cost analysis in the city environment [16]. Although it is a good idea, decentralisation also increases the attack surface, making edge node distributed attacks more likely [6, 18]. Even with the risks being alleviated by the aggregation of fog layers and layered control structures, ensuring secure remote controllability still involves well-coordinated IoT inputs and robust governance structures. Another issue in the fog environment is management overhead which is a persistent challenge in mobile fog systems, where real-time coordination across dynamic nodes is inherently complex. An example is vehicular FC in which vehicles are highly mobile fog

nodes equipped with multiple sensors, which must make decisions quickly, in an aggregated way, and over changing network conditions [17]. Scientists have emphasised the current studies in mobility support and real-time optimisation services [12, 18].

Despite these limitations, real-world deployments demonstrate tangible FC benefits. The SpeziLLM framework, deployed as an iOS healthcare application across six clinical use cases, demonstrates 40% to 60% latency reduction over cloud-based processing with 95% query accuracy [20]. Industrial deployments of Cisco IOx have demonstrated bandwidth savings of up to 80% through edge preprocessing [21]. The VeReMi dataset [22], comprising over 3.15 million vehicular message logs, provides empirical evidence for intrusion detection accuracy of 99.2% in real vehicular fog networks. The classification is synthesized from recurring patterns reported across surveyed literature rather than derived from a single prior taxonomy.

1.6 Classification of FC

Figure 2 illustrates the classification of FC into three primary groups: Fog characteristics, service models, as well as architecture models. The primary function of FC is that it brings the computation closer to where the data originates. Instead of transmitting all data to remote cloud servers, FC lets the processing take place at the source of the data.

1.6.1 Fog Characteristics

The primary differentiators of FC are reflected in its key attributes, particularly edge intelligence. Edge intelligence refers to the capability of systems to perform data processing and decision-making at or near the point of data generation, such as smartphones, sensors, and other IoT devices [23]. Instead of transmitting all data to a distant central data centre, fog-enabled systems process information locally. This proximity significantly reduces data travel distance, leading to lower latency and markedly faster response times.

Furthermore, FC enables geolocation-aware variability, allowing system behaviour to adapt dynamically to where the user is located, for example, adapting dynamically to geographical location. Another distinguishing attribute is real-time analytics, which involves analysing streaming or episodic data as it arrives to produce immediate insights. Additionally, edge intelligence supports mobility support, ensuring that devices like smartphones maintain consistent performance whether stationary or in motion.

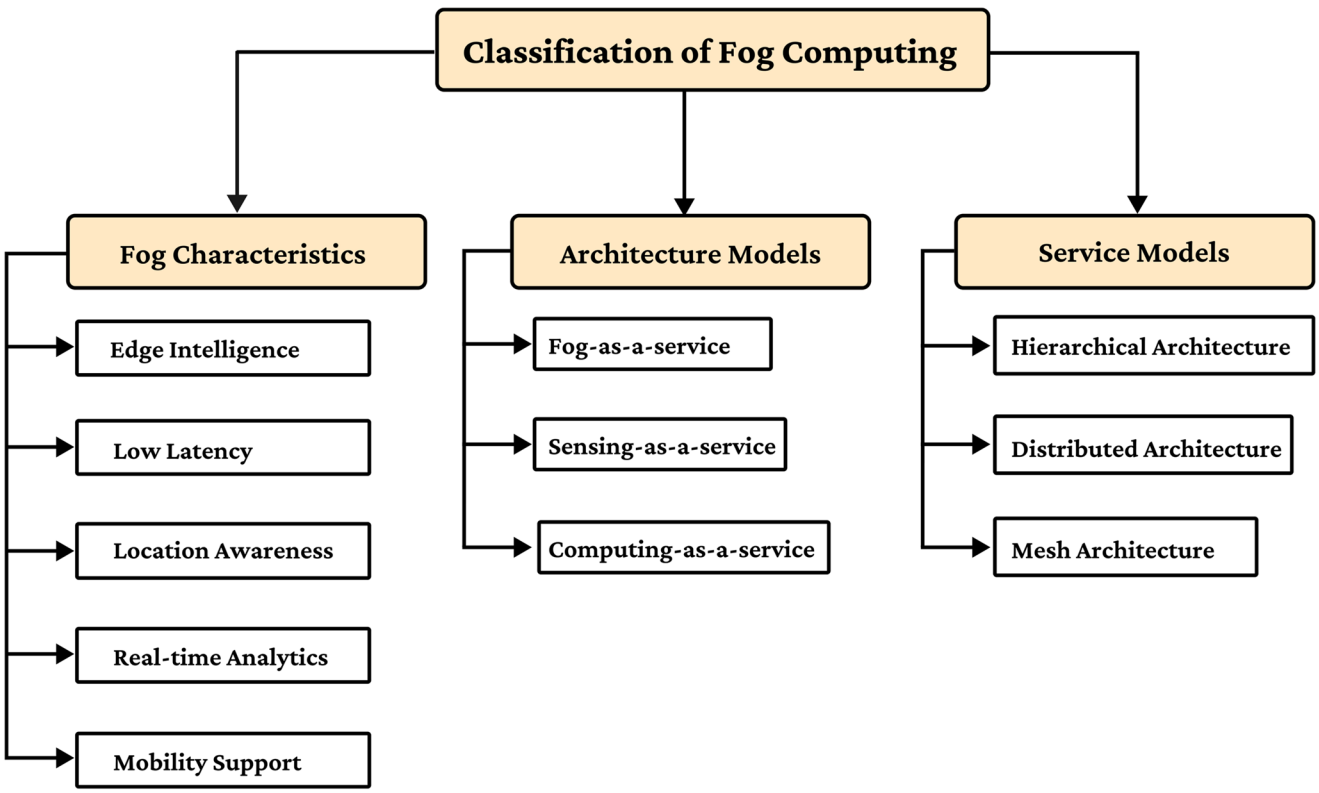


Fig. 2 Classification of fog computing

1.6.2 Service Models

Digital FC offers a range of service models that enhance flexibility in various applications domains. The concept of Fog Infrastructure as a Service (FIaaS) refers to a framework in which all the required fog infrastructure is available and equipment are provided by a service provider, which is why users can access the fog's resources without having to purchase and maintain the hardware. Sensing-as-a-Service (SaaS) utilises distributed sensors to measure and process data at a local level. In the case of a weather station, the station can obtain environmental data and send it in real-time to multiple users. Computing-as-a-Service (CaaS) is also analogous; it provides access to computational resources located close to the source of data, as opposed to being in a remote data centre, which reduces latency and enhances operational efficiency. All these service models expand the scope of applicability of FC and facilitate its application to a broad spectrum of real-world systems.

This survey paper provides a comparative, comprehensive and up-to-date perspective of FC literature, and it finds gaps in previous research. Our distinct contributions are:

- A structured taxonomy is presented classifying FC architectures, service models, scheduling strategies, and security mechanisms across 56 reviewed studies.A comparative review distinguishes FC from cloud and edge paradigms, summarizing performance insights from real-world deployments and simulations.Key challenges including latency, scalability, heterogeneity, and security are comprehensively reviewed and mapped to existing solution approaches from recent literature.AI/ML techniques applied to FC are systematically surveyed and categorized into privacy-focused, energy-focused, and hybrid approaches with comparative analysis.The FogAI-Optimizer is proposed as a conceptual six-layer framework, synthesized from surveyed literature to guide future implementation of intelligent fog systems.

1.7 Novelty of This Survey

Unlike prior FC surveys that address individual dimensions in isolation, for instance [4], focuses on AI-based scheduling but omits green computing [9], covers energy trends but excludes AI integration, and [24] provides a taxonomy without dataset-driven validation. To the best of our knowledge, this survey is among the few existing surveys simultaneously address all six dimensions; most studies focus on two or three aspects. six critical FC dimensions—namely infrastructure architecture, green computing, AI-driven optimisation, cloud–fog integration, edge intelligence, and empirical dataset validation within a unified perspective. This is empirically demonstrated, where no existing survey (2020 to 2025) covers more than three dimensions concurrently. The proposed FogAI Optimizer framework represents a qualitative shift from fragmented, domain-specific studies to an integrated, multi-objective architectural model grounded in 56 systematically reviewed papers. Furthermore, unlike purely conceptual frameworks, FogAI-Optimizer uniquely emphasises the incorporation of real-world healthcare and vehicular datasets for reproducible benchmarking, which has been rarely addressed jointly in prior surveys. This integrative contribution distinguishes our work from incremental extensions and positions it as a structural synthesis of the field.

1.8 Organization of the Paper

Section 2 provides background on FC, covering its historical evolution, key milestones from 2012 to 2025, and identified research gaps. Section 3 critically examines related work across six themes: FC architectures, AI/ML techniques, vehicular FC, task scheduling and optimisation algorithms, security and privacy, domain-specific applications, and energy efficiency. Section 4 outlines the PRISMA-inspired systematic review methodology, including research questions, search strategy, screening phases, and quality assessment criteria. Section 5 reports the review process outcomes, detailing the multi-stage filtering results and characteristics of the 56 final selected studies. Section 6 presents the findings of the review, synthesizing evidence on drivers of advanced FC paradigms, challenges addressed, AI/ML approaches, architectural mechanisms, task scheduling strategies, and practical deployment considerations. Section 7 summarizes the datasets and analytical tools used for fog-based evaluation, including the VeReMi vehicular dataset and Stanford healthcare frameworks. Section 8 proposes the FogAI-Optimizer, a conceptual six-layer integrated framework synthesized from the reviewed literature, along with its implementation roadmap and comparative analysis. Section 9 concludes the paper by summarizing key contributions and outlining future research directions for intelligent, sustainable, and scalable fog computing systems.

1.9 List of Acronyms

Abbreviations and acronyms used throughout this survey are defined in Table 1.

Table 1 List of abbreviations

Abbr.	Full Form	Abbr.	Full Form
5G	Fifth Generation	HVAC	Heating, Ventilation, and Air Conditioning
6G	Sixth Generation	IC	Inclusion Criterion
A-CIDS	Adaptive Collaborative Intrusion Detection System	IEEE	Institute of Electrical and Electronics Engineers
ACM	Association for Computing Machinery	IoE	Internet of Everything
ACO	Ant Colony Optimisation	IoHT	Internet of Health Things
AFedGRL	Asynchronous Federated Graph Reinforcement Learning	IoMT	Internet of Medical Things
AHP	Analytic Hierarchy Process	IoT	Internet of Things
AI	Artificial Intelligence	IoV	Internet of Vehicles
AIoT	Artificial Intelligence of Things	IPAQ	Integrated PSO + AHP
AMQP	Advanced Message Queuing Protocol	ISVR	Improved Support Vector Regression
AMSTAR	A Measurement Tool to Assess Systematic Reviews	ITS	Intelligent Transportation System
API	Application Programming Interface	KMS	Key Management System
AWS	Amazon Web Services	LLM	Large Language Model
BGV	Brakerski–Gentry–Vaikuntanathan Scheme	LSTM	Long Short-Term Memory
BOaaS	Business Outcomes-as-a-Service	LTSFT	Learning-based Task Scheduling with Fault Tolerance
BSM	Basic Safety Message	MADDPG	Multi-Agent Deep Deterministic Policy Gradient
CaaS	Computing-as-a-Service	MADRL	Multi-Agent Deep Reinforcement Learning
CAPEX	Capital Expenditure	MARL	Multi-Agent Reinforcement Learning
CC	Cloud Computing	MEC	Multi-Access Edge Computing
CDN	Content Delivery Network	MIMO	Multiple-Input Multiple-Output
CIDS	Collaborative Intrusion Detection System	ML	Machine Learning
CloudHSM	Cloud Hardware Security Module	MLTPED-BFC	Machine learning-based trust prediction for edge devices in blockchain-enabled fog computing
CPU	Central Processing Unit	MQTT	Message Queuing Telemetry Transport
DAG	Directed Acyclic Graph	NFT	Non-Fungible Token
DCO	Drawer Cosine Optimisation	NLP	Natural Language Processing
DDPG	Deep Deterministic Policy Gradient	OESAS	Optimal Energy and Security-Aware Scheduling
DL	Deep Learning	OPC-UA	Open Platform Communications Unified Architecture
DoS	Denial of Service	PoHV	Probability of Health Vulnerability
DQN	Deep Q-Network	PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
DRL	Deep Reinforcement Learning	PSO	Particle Swarm Optimisation
DROO	Deep Reinforcement Learning based Online Offloading	QoS	Quality of Service
EC	Exclusion Criterion	RABR	Reliability and Bandwidth Aware Routing
EHR	Electronic Health Record	RBCR	Reliability-Bandwidth Constrained Routing
EI	Edge Intelligence	RFID	Radio Frequency Identification
ERO	Efficient Resource Orchestration	RL	Reinforcement Learning
ETSI	European Telecommunications Standards Institute	RNN	Recurrent Neural Network
FaaS	Fog-as-a-Service	RQ	Research Question
FC	Fog Computing	SA	Simulated Annealing
FHE	Fully Homomorphic Encryption	SaaS	Sensing-as-a-Service
FHIR	Fast Healthcare Interoperability Resources	SAC	Soft Actor-Critic
FL	Federated Learning	SDN	Software-Defined Networking
Fog-DaRe	Fog Data Resilience	SIMD	Single Instruction, Multiple Data
GA	Genetic Algorithm	SIoT	Social Internet of Things
GC	Green Computing	SOM	Self-Organizing Map
GHS	Gentry–Halevi–Smart Homomorphic Encryption Optimisation Scheme	SPAWAR	Space and Naval Warfare Systems Command
GPS	Global Positioning System	SVR	Support Vector Regression
GRU	Gated Recurrent Unit	TSM	Temporal Sensitivity Measurement
GWO	Grey Wolf Optimisation	TSSAC	Task Scheduling Strategy with Ant Colony Optimisation
HE	Homomorphic Encryption	UAV	Unmanned Aerial Vehicle

Table 1 (continued)

Abbr.	Full Form	Abbr.	Full Form
HEFT	Heterogeneous Earliest Finish Time	VeReMi	Vehicular Reference Misbehaviour Dataset
HElib	Homomorphic Encryption Library	VFN	Vehicular Fog Network
HETSVP	Hybrid Evolutionary Task Scheduling and VM Placement	VM	Virtual Machine
HIPAA	Health Insurance Portability and Accountability Act	HVAC	Heating Ventilation and Air Conditioning

2 Background

2.1 History and Development of FC

In this subsection, a detailed account of the origins, development and definition of FC is provided. We follow the evolution of it, starting with early distributed computing models to its adoption as a central architecture model in present IoT-based systems. Based on these historical trends, theoretical foundations, and significant technological developments, this discussion establishes the conceptual underpinning that is essential for understanding for understanding the role of FC in contemporary and future studies.

2.2 Historical Context

FC was conceived from several scenarios and did not emerge as a standalone architecture. In the 1990s, companies like Akamai began utilizing Content Delivery Networks (CDNs) by caching content in areas closer to users, which helped improve performance for web interactions; this paved the way for edge processing. Subsequently, in the mid-2000s, CC took root with companies like Amazon Web Services (AWS) and Microsoft Azure, offering centralized processing capabilities for data in data centers. However, with the rapid expansion of IoT devices like smart thermostats, smartwatches, and other wearables and products in the late 2000s, billions of increasingly capable devices were generating data. CC began to show three potential drawbacks: it took too long for responses, especially in real-time applications, CC consumed too much bandwidth, and outages were possible in the cloud's network [13], leading to reports of insecure situations. These challenges necessitated alternative thinking for computation. In 2012, a study [25], introduced the foundational concepts of FC. The main difference indicates that FC is a form of edge computing, but within a hierarchical and flexible architecture, specifically designed for government or multi-level networks.

2.3 Key Milestones

2.3.1 Conceptual Foundations and Early Prototypes (2012–2015)

In this initial phase, Cisco developed the framework, and the Chinese Academy of Sciences began prototyping smart IoT gateways and edge computing devices [26]. In 2014, Cisco created IOx and subsequent versions of IOx, with later versions featuring flexible integration and edge model hyperagents to facilitate specific standards for edge computing in industrial domains. Subsequently, Cisco and Intel participants began working through the OpenFog Consortium's formal standards to define and create standards for interoperability, enabling them to join forces in developing standards to facilitate interoperability [27].

2.3.2 Architecture Development, Standards, and Toolchains (2016–2018)

Researchers began developing their own tools. For example, in 2016, the iFogSim simulator was developed to enable researchers to model early fog systems. The group released the version of IEEE standards in 2018 for identifying fog functionality with cloud 5 G [28]. In 2017, the US Navy's Joint Forces in San Diego adopted some forms of interoperability for fog and other similar IoT technologies [29].

2.3.3 Fog Domain Adoption and Integration Challenges (2019–2021)

While numerous studies have explored the utility of FC in healthcare, the literature on similar IoT and FC mechanisms in other domains like smart cities, agriculture, transportation, and industrial automation is limited [29]. Parallel to the work done in healthcare, several initial investigations also explored FC during the same period. These studies proposed advanced frameworks and demonstrated the use of edge devices like Raspberry Pi for implementing practical fog-based applications in 2019. However, despite these efforts, issues related to data interchangeability within such fog-based systems remain unresolved [30].

2.3.4 Scalable Deployments and Large-Scale Implementations (2022–2025)

Atos introduced Business Outcomes-as-a-Service (BOaaS) in 2022 as a new high-tier service offering [31]. Subsequently, ongoing research in 2024 highlighted the potential of Deep Reinforcement Learning (DRL) for AI-driven task offloading in fog and edge environments [32]. By 2025, more advanced agricultural applications of FC had also emerged [21]. Toward the last quarter of 2025, research trends indicate a convergence of FC with 6 G and AIoT technologies, with increasing emphasis on energy-efficient architectures and the use of trust-based or FL approaches.

Figure 3 illustrates the phased evolution of FC across these key milestones, highlighting the transition from early conceptual foundations to large-scale, intelligent deployments.

Here, modern FC is critical for addressing modern computing demands placed on computing workloads. For example, the practicality of a self-driving car requires processing sensor data in milliseconds, which cannot be guaranteed with a cloud server due to distance and latency (typically more than 100–500 ms). By utilising fog nodes, the overall latency can be reduced by more than fivefold with processing of data handled locally at sub-10 ms latency [3]. In addition to decreased latency with fog systems, there is a decrease in bandwidth costs for sending data when it transmits only 20% of the original data volume from the edge. Cisco IOx has demonstrated bandwidth savings as high as 80% [24]. Another key aspect of FC is that the data is secure, as we only transmit that which is necessary when moving data from edge devices to the main cloud service or to another cloud service [25].

The paradigm shift from cloud to fog is comprehensively characterized in [13], which identifies interoperability and cost-effectiveness as the twin determinants of mainstream industrial fog adoption.

3 Related Previous Work

While several FC surveys exist, this section deliberately positions our contributions against them. Unlike [4] and [17], which catalogue RL and deep learning variants at a high level without quantified per application benefits, our AI taxonomy explicitly distinguishes privacy-focused (FL, homomorphic encryption), energy focused (DRL, evolutionary algorithms), and hybrid AI/ML approaches each with specific smart city use cases and measurable performance gains. Unlike [9], which provides a decade-level retrospective without architectural synthesis, we construct a six-layer integrated framework (FogAI-Optimizer)

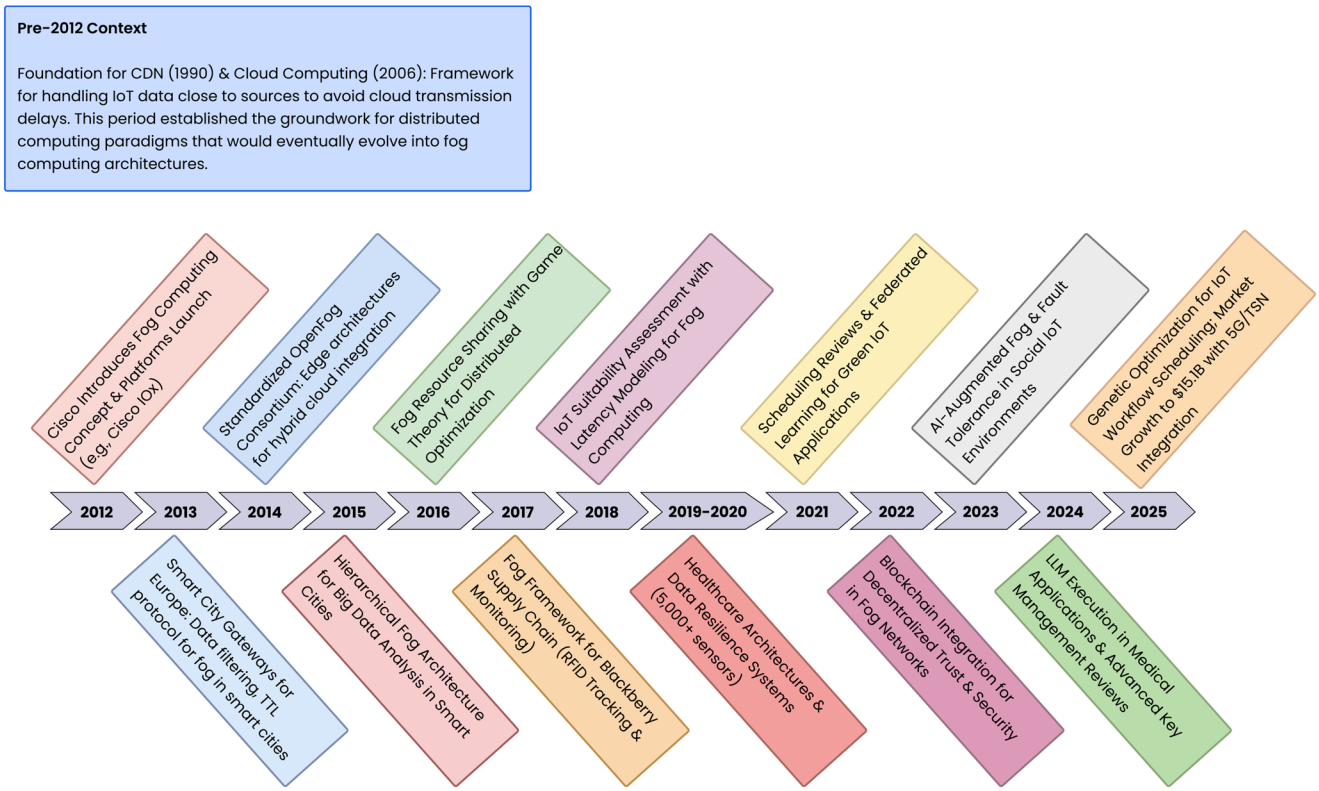


Fig. 3 Evolution timeline of fog computing

grounded in 56 reviewed studies. Unlike [24], which offers a taxonomy without implementation guidance, we provide a phased implementation roadmap (Sect. 8.4) and integrate eight real world datasets for reproducible validation (Table 11). These distinctions collectively justify the need for this survey beyond existing literature.

3.1 Fundamentals and Architecture of FC

Next-generation IoT applications generate massive volumes of data; however, centralized processing in CC introduces latency and energy overheads for time-critical applications. This comprehensive survey delves into FC, including layered design decisions, latency reduction goals, resource management, and workload distribution methods through resource management schemes. Each application is evaluated and analyzed accordingly, with in-depth analyses of smart healthcare and vehicular applications, indicating that FC can reduce latency by 50% to 70% compared to CC, however, ensuring security in heterogeneous fog environments remains a challenge [14, 33]. Critical fast and secure IoT applications are essential in smart cities, particularly for emergency response and health monitoring [34]. CC is unsuitable for latency-sensitive smart city applications due to latency and security limitations. The research examines FC approaches for smart cities in three contexts: reducing latency while providing services, conserving energy while providing services, and integrating urban IoT focused on applications in smart cities. Evaluation using simulation tools like iFogSim shows that FC approach reduces latency by 50% to 70% compared to CC for urban applications, even with high application complexity. The multidimensional complexity of resource management and maintaining ongoing security objectives when scaling systems based on FC models is a persistent problem and has been extensively described in the literature [35, 36].

3.2 Artificial Intelligence and Machine Learning in Fog Computing

Unlike prior AI-in-fog surveys [37, 38] that catalogue RL and DL variants at a high level, this survey's AI taxonomy distinguishes privacy-focused, energy-focused, and hybrid AI/ML approaches with specific use cases and quantified benefits [39] (Table 5).

3.2.1 Reinforcement Learning for Resource Management

Resource management within FC presents complex optimisation problems that traditional optimisation methods struggle to solve due to the dynamic and heterogeneous characteristics of the system. Existing studies (2015–2025) explore RL-based resource management approaches in FC, examining resource provisioning, task placement, job scheduling, resource allocation, computational offloading, and load balancing. The classification highlights the relationships between RL variants exemplified by deep Q-Networks and Deep Deterministic Policy Gradient (DDPG) and the challenges specifically associated with FC. Compared to heuristic-based approaches, RL-based approaches report 20% to 40% improvements in latency and energy efficiency across multiple studies. However, challenges exist in training and scaling RL techniques. Multi-agent RL shows promise for application in large-scale deployments, with future work focuses on integrating RL with FL [40].

The rapid evolution of distributed computing paradigms will necessitate the application of certain AI capabilities to manage them effectively [23]. In particular, there has been little evaluation of the impact on quality of service and resource utilization, addressing stability mechanisms, and leveraging functionalities available with Moore's Law [16].

3.2.2 Deep Learning for Task Offloading and Scheduling

Task offloading in FC faces challenges when dealing with dynamic changes in topology and heterogeneous devices. Asynchronous Federated Graph Reinforcement Learning (AFedGRL) combines two techniques: (1) Reinforcement Learning with actor-critic algorithm for topology-aware task offloading decisions, and (2) Graph Neural Networks (GNN) to model network topology and inter-device relationships. To support these improvements, it leverages the continuous evolution of traditional FL systems, allowing the system to operate asynchronously [41], and therefore removing synchronization constraints among devices. Through an evaluation model compared to DROO and HEFT, the offloading gain achieved through AFedGRL is 15% higher than other models, while latency is reduced by 25%, and completion rates are 13% higher. AFedGRL enables continuous model updates under asynchronous network conditions, it helps active devices continuously learn without interruption. FC essentially distributes IoT processing and reduces resource dependency on the cloud for use cases associated with critical applications for instance healthcare monitoring or industrial automation; however, users are left to manage resources based on the stochastic dynamics of task arrival in the system, which introduces latency issues in environmentally context-dependent scenarios. Envisioned is a framework built upon DDPG concepts, realizing adaptive scheduling of resource offloading, task scheduling, and adaptive resource allocation considerations for sequential, parallel, and dependency-aware scheduling frameworks. By supporting various scheduling types and handling continuous action locations, Google Cloud job simulation has

demonstrated improvements of up to 30% in completion time and fault tolerance, a 40% reduction in bandwidth and latency, and a 20% increase in reliability for complex IoT workloads [42].

Recent advances have extended task offloading to Soft Actor-Critic (SAC) algorithms, which introduce entropy regularization enabling superior exploration stability in non-stationary fog environments. Moghaddasi and Jurdak [43] have proposed an energy-aware distributed federated SAC framework for intelligent task offloading in vehicular mobile edge computing networks. Their framework achieves from 18% to 22% better energy efficiency than DDPG-based methods under dynamic vehicular channel conditions, while maintaining data privacy through federated model updates that eliminate raw data exchange between vehicles and infrastructure. The federated architecture enables continuous learning without centralized aggregation, overcoming a key limitation of synchronous FL.

3.2.3 Machine Learning for User Classification and Resource Allocation

The expanding deployment of IoT saturates CC with latency and bandwidth issues related to mobility, while traditional allocation is inefficient for real-time dynamic mobile fog environments. Classifying users based on movement patterns through ML clustering with K-means or Self-Organizing Maps (SOM), and combining this with FC, or edge-based processing, can reduce energy consumption through proximity-based allocation. SOM achieves 10% to 15% higher clustering accuracy on average and clustering quality compared to K-means, and both approaches yield from 20% to 30% energy consumption and 25% latency savings over random allocation, enabling scalable fog deployments for augmented reality and vehicular applications. Systems are limited by bandwidth and latency constraints in mobile settings, as well as the inability of traditional allocation approaches to properly classify dynamic users, resulting in inefficient resource utilization. Data classification using ML clustering techniques includes K-means clustering for partitioning and SOM for clustering, which combines a linear update of centroids with topology-preserving classification of data clustering users based on mobility and proximity patterns [37].

3.3 Vehicular FC and Real-Time Applications

3.3.1 Vehicular Network Resource Management

Vehicular networks provide low latency between cloud resources, task processing through fog access points, and machine-to-machine or vehicle-to-vehicle communication. Furthermore, because different vehicles have varying workloads and hardware capabilities, the mobility of vehicles means that the systems must support heterogeneous vehicular capabilities without requiring changes in the fog at the edge [44]. As such, cell-free massive Multiple-Input Multiple-Output (MIMO) technology enables the use of multiple fog access points and allows a vehicle to access them without establishing a handover. Typically, Long Short-Term Memory (LSTM) networks can be employed to predict the workload at the fog access points. The methodology also utilises Multi-Agent Deep Deterministic Policy Gradient (MADDPG) RL to optimize resource allocation and task distribution. The approach reduces both energy consumption and latency from 15% to 25%. With a 20% improvement in latency compared to existing methods, this approach has the greatest impact in areas with high vehicle density. Limited fog node and roadside unit resources in vehicular networks lead to limited performance during peak demand periods. The Efficient Resource Orchestration (ERO) algorithm logically divides fog coverage into restricted and unrestricted areas of the road and redistributes tasks according to capacity utilization based on priorities when tasks are assigned. Load balancing prevents resource overload at the individual node level. The results indicate a 20% to 30% increased performance, a 15% improvement in service class abstraction capability, and a general capacity that is 10% to 20% higher than existing methods, especially under heavy traffic conditions. The continuous movement of vehicles in a vehicular FC environment creates low latency as service tasks are being scheduled and executed, leading to high energy consumption due to service delays. The mathematical process model describes the vehicles' movements, while the DRL model is used to employ an adaptive approach to optimize scheduling and resource allocation. Collaboration with other roadside units increases the ability to effectively share workloads and resources. Testing in an urban environment resulted in an average reduction of 12% in service delays and 14% in energy consumption compared to other more basic methods available [45].

3.3.2 Reliability and Anomaly Detection

Advanced Gated Recurrent Unit (GRU)-Autoencoder for reliability anomaly detection in vehicular FC Services. Ensuring the reliability of FC services operating in vehicular environments, considering factors like battery power limitations, memory usage, and network congestion, is a significant challenge. Currently, available anomaly detection systems primarily focus on detecting security threats rather than service reliability issues. The proposed improved GRU-Autoencoder model aims to extract normal service behavior patterns while identifying anomalous behavior patterns. The

process of determining the optimal threshold for detecting anomalous behavior patterns is conducted using Bayesian Optimisation. During the training process, a multi-stage scored reservoir sampling method is used to select the most representative samples of the training data. A checkpoint-restart mechanism allows for rapid recovery to a stable state after anomaly detection. Tests conducted on five vehicular scenarios demonstrate detection accuracy between 95% to 98%. The improved model is able to identify patterns that were not previously detected by earlier autoencoder models, with an improvement of approximately from 10% to 15%, and also achieves reduced downtime through faster recovery to a normal state [46].

3.3.3 Autonomous Driving and Task Offloading

Autonomous vehicles experience delays due to limited edge–fog resources, while existing offloading mechanisms neglect to address sub-tasks and additional dependency issues and cache management [44]. Formulating task scheduling and cache updating expressions as optimisation problems allows for considering vehicle-to-vehicle or vehicle-to-infrastructure or local submission methods. The developed discrete cuckoo search algorithm demonstrate improvement in exploration with Lévy flights and adaptive probability for convergence, thereby enhancing offloading efficiency. The results showed a 12% increase in offloading efficiency. The approach reduces latency, increases throughput, and decreases cache misses by 15

3.3.4 Cooperative Vehicle Sensing

Cooperative vehicular network sensing results in increased demands on the sensing, spectrum, and computational resources available for this research; this directly impacts applications requiring safety-critical responses for autonomous vehicle systems. Current approaches lack a method for balancing sensing accuracy, communication efficiency, and computation processing time. This recommendation generates cluster-based solutions for vehicle clustering, message selection, and radio-computing resource allocation that are jointly optimized; through student-specific and attention-enhanced multi-agent DRL for clustering and selection, and applied swap matching algorithms, as well as interior-point methods for resource optimisation. Test results in adjacent cluster-based vehicle processing demonstrate a 15% to 20% improvement in sensing accuracy, reduced latency and energy consumption from 10% to 25%, and a higher overall task completion rate compared to random clustering methods. The performance gains are particularly significant in densely populated urban environments [47].

3.3.5 Smart Traffic and Internet of Vehicles

Vehicular systems generate numerous time-sensitive smart traffic tasks, but processing workloads solely on the cloud introduces delays [34]. Static FC mechanisms and threshold-based vehicular fog node models are insensitive to mobile operations, potentially exacerbating resource imbalances and high service latency. This model introduces a hybrid cloud and vehicular FC approach, channeling mobile fog nodes through optimisation for minimum acceptable latency using bifurcation methods and clustering. It enhances the acquisition of vehicular service requests through a joint optimisation approach during high mobility and addresses the structural fracturing of user-designated vehicular fog nodes. The algorithm is capable of making decisions based on the mobility of both nodes and vehicles. The results show a 25% to 35% reduction in maximum acceptable delay compared to processing workloads using only cloud infrastructure, while also demonstrating a 20% improvement in fairness for all users and a 15% increase in queuing efficiency for all reviewed vehicle-to-vehicle applications [48].

3.3.6 Blockchain-Enabled Security in 5 G Vehicular

Fog networks beyond performance optimisation, security and trust in vehicular FC environments present distinct challenges in 5 G-enabled networks. Blockchain-based approaches have been proposed for secure inter-vehicle data sharing in fog-assisted 5 G architectures, where the decentralized ledger ensures data integrity while fog nodes provide low-latency local validation [49]. This approach eliminates single points of failure common in centralized trust models, achieving verifiable data provenance with minimal computational overhead on vehicular fog nodes. When combined with FL mechanisms [43, 50], blockchain-enabled vehicular fog systems represent a promising convergence of security, privacy, and performance optimisation for next-generation intelligent transportation systems.

3.4 Task Scheduling and Optimisation Algorithms

3.4.1 Fault-Tolerant Scheduling

FC frameworks allow IoT-enabled organizations to achieve low-latency data processing; however, they must address reliability concerns surrounding node failures, for instance heterogeneous hardware configurations, fluctuating resource availability, and unreliable power supply. For fault handling, the proposed Learning-based Task Scheduling with Fault Tolerance (LTSFT) policy utilises variable-structure learning automata to enable intelligent decision-making. The LTSFT technique also predicts potential failures proactively,

allowing the execution of the original task on another available node or the concurrent execution of tasks across multiple nodes during task execution. This policy adapts itself based on surrounding conditions and determines the best fault management approach by estimating the total time and energy consumed during task execution [51]. In experimental validation, the LTSFT algorithm demonstrated a 20% to 30% improvement in the quality of service for task execution, a 15% to 25% reduction in total energy consumption, and a 10% to 20% increase in overall resource utilization.

3.4.2 Multi-Objective Optimisation

Inadequate multi-objective optimisation in fog task offloading leads to network congestion and processing delays. Drawer Cosine Optimisation (DCO) addresses this by combining the Drawer algorithm with a sine cosine algorithm, providing offloading decisions that simultaneously optimize completion time, cost, workload distribution, and energy consumption. The master node uses DCO for optimal scheduling. Tests conducted with 100 tasks identified a load factor of 0.116, energy consumption of 0.472J, a completion ratio of 0.365, a processing time of 3.221 seconds, and memory usage of 7.452MB. The method achieves 10% to 20% performance improvement compared with baseline approaches [52].

3.4.3 Fuzzy Logic–Based Scheduling

FC also offers reduced latency in processing the delay factor introduced by IoT, based on the volume of data and time-sensitive processes. However, managing large volumes of time-sensitive data strains the parameters of existing resources, resulting in longer execution times, wasted energy consumption, and missed deadlines, none of which can be resolved by CC environments. To address this with IoT workloads, a semi-greedy algorithm based on fuzzy logic (FuzzyPSG-M) is proposed for adaptive decision-making in uncertain conditions. A modified multi-start-based framework system was tested, first with a baseline method of classifying each task based on time-sensitive priority, and then applying fuzzy logic rules to provide flexible fog node scheduling. Adding the multi-start option leads to a better exploration of solutions. For testing, two scenarios were used with 300 and 500 tasks on 50 to 60 fog nodes. The two test scenarios showed that FuzzyPSG-M significantly reduces completion time from 10 to 15%, energy requirements by 5% to 10%, and improves deadline satisfaction by approximately 8% to 12% compared to other baseline methods, clearly demonstrating that fuzzy logic-based dynamic task scheduling is effective for FC-related tasks [53].

3.4.4 Meta-Heuristic Approaches

FC can be beneficial when IoT systems require low latency through proximity processing, but the complexity of diverse node capabilities and multiple quality-of-service elements can complicate device placement, lead to the use of unnecessary resources, and incur high costs if not accurately estimated [35]. A meta-heuristic algorithm, inspired by the hunting behavior of Harris's hawks and developed for cost-aware placement, can monitor quality-of-service metrics and node capabilities for deployment, considering some combination of Optimisation in queues, node throughput, energy, and costs. Testing indicates a 4.5% increase in resource utilization, a 3.8% improvement in acceptance rate, a 2.95% reduction in latency, and a 1.62% decrease in energy consumption compared to current solutions, all of which suggest scalability for implementations involving billions of devices [54].

3.4.5 Hybrid Evolutionary Algorithms

Fog systems supporting IoT applications like smart healthcare require reliable task scheduling and virtual machine placement strategies. However, the heterogeneity of cloud-enabled hardware increases completion time and creates energy imbalances. The Hybrid Evolutionary Task Scheduling and Virtual Machine Placement (HETSVP) framework utilises improved particle swarm optimisation with binary encoding and adaptive parameters, along with resource-aware allocation of new virtual machines based on physical machines [55]. The iFogSim evaluation of the HETSVP framework yielded 11% to 13% improvement in completion time, 5% to 15% energy reduction, and 10% to 20% load balancing improvement over minimum minimum, maximum minimum, genetic algorithm, and ant colony optimisation applications, confirming performance improvements in real-time IoT use cases [56].

3.4.6 Ant Colony Optimisation for Industrial IoT

FC for industrial IoT, capable of utilizing large datasets through interdependent tasks represented as directed acyclic graphs, faces challenges due to the wide variation in capabilities of computing hardware nodes allocated to tasks, resulting in longer completion times, workload imbalance, and increased energy expenditure. Standard techniques fail to accommodate the overload or underutilization of resources. Task Scheduling Strategy with Ant Colony Optimisation (TSSAC) is introduced, which utilises ant colony optimisation for completion time, workload, and energy balancing for Directed Acyclic Graph (DAG) tasks. TSSAC employs pheromone factors to guide the ants. The ants

follow a geographical factor, promoting initial exploration that leads to subsequent convergence on an optimal path, which also facilitates resource migration for utilised containers while minimizing task execution overhead and effectively managing resources. Initial test results indicate that TSSAC produced a 23.5% reduction in energy expenditure with improved load balancing and completion time compared to baseline optimisation, while the energy consumption allocation for container migration is applied to produce an estimated 20% load reduction for customer use cases and smart manufacturing [57].

3.4.7 Fairness-Aware Scheduling

The increasing volume of IoT data in FC demands precise scheduling solutions; existing solutions often overlook the differences in time sensitivity and fairness among various types of tasks, potentially leading to unacceptable service and response times. The Integrated PSO+AHP (IPAQ) algorithm have proposed a fair solution that groups tasks together based on time sensitivity using a high-priority-first approach. It incorporates an improved particle swarm optimisation (PSO) technique to provide the task system with the ability to search globally and includes the analytical hierarchy process (AHP) to examine multiple criteria related to latency and quality of service. The integrated system is capable of scheduling and evaluating dynamically in real time. The authors have reported a 15% to 25% reduction in waiting time and response time and a 20% increase in quality of service compared to baseline algorithms, demonstrating added robustness in scheduling tasks across varying sensitivity levels spanning diverse IoT scenarios, including smart cities [58].

3.5 Security and Privacy in FC

3.5.1 AI-Based Security Solutions

FC environments face unique security and privacy threats associated with limited resources and their distributed nature. A systematic PRISMA framework review of over 56 studies from 2015 to 2025 evaluated ML for anomaly detection, FL for privacy preservation, and lightweight AI for devices operating in resource-constrained environments. The classification describes specific AI security solutions across three categories: resource management, security enhancement, and privacy preservation. While AI methods improve average threat detection accuracy by 25% to 35% for the scenarios measured, further development of lightweight and optimized models is needed to address edge-device deployment challenges. FL achieves accuracy exceeding 90% while effectively preserving privacy [59].

3.5.2 Intrusion Detection Systems

Vehicular fog network systems are susceptible to distributed denial-of-service attacks and time-sensitive breaches, yet existing systems lack effectiveness in terms of speed, accuracy, or scalability in their collaborative detection methods. The authors of Adaptive Collaborative Intrusion Detection System (A-CIDS) have proposed a system using ML with a variational autoencoder for feature extraction, an extended isolation forest algorithm for anomaly detection, and a modified synchronous parallel approach for sharing threat intelligence among nodes. The A-CIDS system demonstrated performance metrics of 99.2% accuracy with a 2% false alarm rate and a 150-millisecond response time, which is 10% to 20% outperforming previous systems [60].

3.5.3 Trust Management and Blockchain Integration

In blockchain-enabled fog networks, the proliferation of edge devices and potential trust issues arising from security breaches are complicated by inefficient prediction levels of trustworthiness. The Machine learning-based trust prediction for edge devices in the blockchain enabled fog computing (MLTPED-BFC) approach applies support vector regression with multivariable logistic regression to predict trust scores, and then the trust score update is completed through the fog server. During the prediction process, a trust update is completed for each transaction after the initial update is completed, without any bias towards any individual device. In testing, an accuracy of 98.91%, a precision of 98.92%, a recall of 98.32%, and an F-measure of 98.96% were achieved with a minimum error rate of less than 5%. The metrics also outperformed other methods by a level of 5% to 10% [61]. Centralized models of evidence management within FC continue to pose problems regarding trust and transparency, where data is lost or manipulated through a single point of failure. The challenge facing traditional systems is that they cannot provide the levels of trust required when presented forensically in relation to legal applications or the integrity of evidence. Decentralized systems build a unique foundation that allows for the inclusion of NFTs, blockchain components, and fog infrastructure [41]. NFTs provide unique digital certificates for any digital asset, while the blockchain allows for the functionality of an immutable record after creation. Fog nodes provide low-latency access without exhibiting the compromised form of a centralized system. The record storage methods are immutable, which enhances record integrity, and any attempts at modification can be instantly detected through verification using cryptography [62]. Testing shows a 30% improvement in security performance compared to centralized systems, with a significantly lower likelihood of compromising

the integrity of systems essential for legal proceedings and forensic investigations [63].

3.5.4 Authentication and Access Control

FC also presents implications for node authentication and workload balancing, in that network latency leads to increased energy consumption, security vulnerabilities, and delays for edge data centers. An adaptive probabilistic neural network offers an approach for secure authentication, while a hybrid heap African buffalo optimisation algorithm calibrates parameters and allocates workstation resources considering cost and energy expenditure. This enables greater accuracy, precision, and negative predictive value, while significantly reducing false positives, energy consumption, and cost, making it superior to traditional methods in terms of speed and security [64].

3.5.5 Routing Security in SDN-Based Fog Systems

Software-Defined Networking (SDN) based FC for IoT is susceptible to link failures and network congestion. Achieving the required reliability while simultaneously meeting bandwidth demands clearly highlights the challenges and trade-offs involved. To address this, Reliability and Bandwidth Aware Routing (RABR) and Reliability-Bandwidth Constrained Routing (RBCR) employ innovative and efficient path selection techniques, along with duplication, to route data flows using parallel, hybrid, and sequential approaches. Each routing and duplication stage is evaluated as a separate phase for effective performance assessment. Test results indicated a 15% to 25% improvement in reliability, along with a 20% increased success rate and reduced processing time compared to scheduling by existing SDN-IoT routing methods [20].

3.5.6 Cloud–Fog Key Management and Cryptography

The increased scalability of CC introduces new privacy and security risks, while cloud storage key management services and solutions continue to lack user-based analytics and fail to meet latency and compliance requirements, hindering user adoption. Accordingly, a systematic literature review of 56 academic and peer-reviewed studies published between 2015 to 2025 was conducted using PRISMA guidelines, and a taxonomy was developed to classify key management systems used for cryptographic and architectural approaches. The review also analyzed the use of these tools, like AWS KMS and CloudHSM. Systems experience up to 25% fewer security incidents. Latency is an issue for 40% to 50% of non-interactive workloads and 35% of interactive workloads, but only 20% of implementations. The taxonomy led to a 30% improvement in the selection of key management solutions and systems for enterprise tool usage and compliance implementations [53].

3.6 Domain-Specific Applications

3.6.1 Smart Healthcare

Existing remote pet-health monitoring systems lack mechanisms for detecting slowly developing conditions, for instance cushing's syndrome, that affect pets in domestic settings. Internet of Medical Things (IoMT) devices have limited processing latency or constraints regarding continuous health monitoring. Smart VetCare presents a FC framework for temporal artificial neural networks that analyzes health information over long-term assessments. Novel metrics, Probability of Health Vulnerability (PoHV) and Temporal Sensitivity Measurement (TSM), help predict the likelihood of health problems developing in severity, while SOM provide visualizations of health patterns for veterinarians, validated on real-world veterinary datasets, demonstrating an accuracy of 94.78%, a precision of 95.38%, a sensitivity of 93.71%, and an F-measure of 94.41% in metric evaluation. In comparative isolation, Smart VetCare demonstrated an approximately 5% to 10% increase in diagnosing health problems compared to Recurrent Neural Networks (RNN) and LSTM methodologies. Furthermore, the proposed framework identifies false alerts while allowing for proactive veterinary care [65]. The use of applications powered by large language models to process healthcare data obtained from electronic health records and wearables presents privacy concerns, reduces patient trust, and increases costs due to cloud deployment. The research has suggested a decentralized FC model that utilises edge devices as fog nodes acting as hosts, combined with local fog layers to redistribute the Large Language Model (LLM) processing workload with open-weight models. SpeziLLM provides an open-source, customizable iOS framework to enable dynamic LLM deployment with FHIR-compliant interoperability. Testing across six healthcare applications demonstrated a 40% to 60% reduction in latency compared to cloud-based processing, up to a 70% reduction in costs, 95% accuracy with real-time data queries, and the ability to retain health data within the local device [66].

3.6.2 Industrial IoT and Industry 4.0

Industry 4.0 requires processing high-frequency Industrial IoT data from sensors for AI applications at the edge, fog, or cloud layers [67]. Protocols like Message Queuing Telemetry Transport (MQTT) or Advanced Message Queuing Protocol (AMQP) suffer from processing latency and lack

of timely event handling for centralized processing, and five protocols, including AMQP, MQTT, Kafka, ZeroMQ, and Open Platform Communications Unified Architecture (OPCUA), are benchmarked for high-throughput transmission to evaluate latency and stability in computing architectures. MQTT, AMQP, and ZeroMQ are identified as the best protocols for variations under 10 milliseconds, and ZeroMQ is identified as the best protocol for the cloud with latency under 5 milliseconds, and can increase system efficiency in industrial applications by 20% to 30%, for example, in predictive maintenance [68].

3.7 Energy Efficiency and Green FC

3.7.1 Green AI Techniques

Without sustainability considerations, fog and IoT-based AI consume a significant amount of energy and processing power. Through a research survey of green AI techniques, it becomes clear that AI models are more efficient for edge and fog applications through pruning and quantization [55]. Investigating IoT-specific implementations for low-energy AI systems has become crucial. This leads to the development of efficient AI models and reduce energy consumption for fog AI applications by 30% to 50%. However, careful consideration of the trade-off between accuracy and efficiency will always be necessary [69].

3.7.2 Green IoT Architectures

As billions of IoT devices increase energy consumption, carbon footprint, and electronic waste, some traditional designs are incorporating sustainability considerations for use cases like smart climate control in smart cities and buildings. Green IoT architecture utilises existing renewable energy sources, considers the role of low-power sensor technologies, analyzes applications related to Heating Ventilation and Air Conditioning (HVAC) integration, and provides a renewable energy grid architecture for computational offloading [67]. The Green IoT architecture aims to reduce energy consumption by 30% to 50% through the application of edge processing [54, 67]. The architecture also reduces HVAC energy load by up to 25%, significantly contributing to closing the sustainability gap at the application layer [70].

3.7.3 Sustainable Supply Chain Management

The goal of reducing carbon emissions faces obstacles because traditional supply chain systems do not optimize environmental parameters when AI-IoT is not integrated into the optimisation methods. An AI-IoT framework for green supply chains, utilizing Improved Support Vector Regression (ISVR) for carbon emission forecasting, will consider green management practices, eco-friendly product selection, and clean technology in three areas: transportation, manufacturing, and waste management. Achieving net-zero carbon emissions will require optimisation across these three areas. According to simulation results, emissions were reduced by more than 20% to 30% compared to traditional practices, while still maintaining 92% accuracy and a measurable environmental impact on operations within Industry 4.0 manufacturing [71].

Table 2 provides focus area captures the primary research theme, like fault tolerance, data resilience, energy efficiency, or task optimisation, while the key contribution highlights each study's main intellectual advancement. The differentiation from our survey, clarifies how these studies diverge from the present work, particularly regarding breadth of coverage, architectural generality, or integration of AI-driven perspectives.

3.8 Research Gaps

- Initial implementations focused on latency to meet performance requirements; recent surveys have appropriately addressed the lack of power consumption studies [21].
- Security not only has the use of edge computing devices increased, but so have the vulnerabilities arising from the use of IoT devices [25].
- As we look towards the 6 G future, scaling resources on fog services will continue to challenge manageable systems from hundreds of thousands of nodes to millions of nodes [32].

4 Review Methodology

This study follows an adapted PRISMA methodology tailored for computer science literature. Standard PRISMA guidelines were developed for biomedical systematic reviews and require domain-specific adaptation for computer science research. Our adaptations include: (1) inclusion of top-tier conference proceedings alongside journal articles; (2) Quality assessment criteria tailored to algorithmic and simulation-based research; (3) broader search scope including IEEE Xplore, ACM DL, and engineering databases not standard in biomedical PRISMA. The multi-stage filtering process follows the PRISMA Identification–Screening–Eligibility–Inclusion workflow: 2,627 records were identified, 742 remained after title screening, 454 after

Table 2 Comparison of related work across FC and AI domains

Ref.	Year	Focus Area	Key Contribution	Methodology	Strength	Limitations	Differentiation from Our Survey
[72]	2020	AI applications in disaster management	Comprehensive review of AI for disaster response	Literature review	Covers all disaster phases	Response phase bias	Disaster focus; ours targets general fog AI applications
[73]	2021	Data resilience in FC	Fog-DaRe system for data persistence and compression	Filtering, store-and-forward	Reduces storage and transfer by 77% to 83%	Simulated sensors only	Resilience focus; ours emphasises AI-driven enhancements
[74]	2018	AI-based fog and edge computing review	Taxonomy and future directions for AI in fog/edge	Systematic literature review	Comprehensive AI trends	High-level, no implementation	Review-based; ours includes practical AI applications
[75]	2023	Fault tolerance in fog-based SIoT	Fault-tolerant clustering with Markov chains	Clustering, Markov modelling	Enhances reliability	SIoT-specific	SIoT focus; ours extends to general fog reliability
[38]	2024	AI-augmented edge and fog trends	Trends and challenges in AI for edge/fog resource management	Systematic review	Highlights scalability challenges	Conceptual only	Trend analysis; ours includes specific AI implementations
[76]	2024	Sustainable computing in fog-cloud	Holistic framework for energy-efficient computing	Case study, framework design	Real-world applicability	Context-specific	Sustainability focus; ours integrates AI innovations
[39]	2023	Energy-efficient resource generation	PowerGen framework for edge-cloud resource utilisation	Data generation framework	Supports energy-aware scheduling	Synthetic data	Energy focus; ours broadens to AI-driven resource use
[77]	2025	Dynamic fog for LLM in healthcare	SpeziLLM framework for local LLM execution in medical apps	Open-source framework, fog deployment	Enhances privacy and trust	Healthcare-specific	Medical focus; ours generalises AI in fog
[78]	2023	Decentralized trust in FC	NFT and blockchain-enabled evidence system	Blockchain integration, fog processing	Improves data integrity	Blockchain-specific	Security focus; ours includes broader AI integration
[79]	2025	Key management in cloud-fog hybrids	Systematic review of key management for security	Systematic literature review	Identifies security gaps	High-level review	Security review; ours focuses on AI-driven fog solutions
[80]	2021	Deadline-constrained IoT workflow scheduling	Genetic algorithm with selective repair for IoT workflows under deadlines	Genetic algorithm, selective repair	Optimises cost with deadline adherence	Synthetic workflows	Focuses on genetic Optimisation; ours integrates broader AI

abstract screening, 210 after introduction and conclusion review, and 56 studies were finally included.

In this section, we present the methodological framework used to review the extensive body of work on FC spanning the years 2015 to 2025, informed by a collection of systematic literature reviews that systematically categorize IoT-fog architectures to support smart surveillance, and identify gaps in the literature regarding the use of FC technologies for proactive policing, among other scenarios. Our methodology facilitates a comprehensive and reproducible review of any discussed thematic categories of the literature on FC: task scheduling, resource allocation, energy efficiency, and AI/ML. This process adheres to the PRISMA principles of inclusion and reporting, while emphasizing the identification of gaps in the descriptions of FC paradigms to enable readers with subject-matter expertise and general audiences to engage with the material in a transparent and easily understandable manner.

4.1 Research Objectives

This survey is a comprehensive FC review with focused analysis of three high-impact subdomains: AI/ML (IC-3), green computing (IC-4), and vehicular fog/MEC (IC-5). These were selected because they represent the dominant contemporary research trajectories. However, foundational fog studies addressing architectures, algorithms, and IoT applications (IC-1) are included regardless of AI, green, or vehicular. IC-3, IC-4, and IC-5 are evaluated with OR logic: a paper addressing ANY ONE qualifies alongside IC-1.

The objectives are based on themes that were recurring in the papers contributing to the collection, aiming to systematically identify research gaps in FC.

4.2 Research Questions

This structured SLR will focus on the significant challenges, solutions, and applications of FC in IoT applications. As stated in the review and to achieve the research objectives outlined above, the following research questions (RQs) were formulated to guide and support the analysis:

- RQ1: What factors motivate the design of novel FC paradigms in IoT environments?
- RQ2: What problems are addressed by FC solutions?
- RQ3: What privacy-preserving and energy-efficient AI/ML approaches are applied to FC in dynamic environments?
- RQ4: What architectural models exist in FC?
- RQ5: What challenges and solutions exist in FC task scheduling and resource management?

The investigation of the role of FC is focused on the challenges of FC, solutions, and research gaps.

4.3 Search and Selection Strategy

The rapidly evolving paradigm of FC prompted us to adopt a survey strategy that required us to empirically collect relevant works from all scholarly sources. Thus, we developed a research-based knowledge base by gathering material from: "ACM Digital Library," "arXiv," "IEEE Xplore," "ScienceDirect," "SpringerLink," and "Wiley Online Library".

4.3.1 Search String

(Fog Computing OR Edge-fog-cloud continuum OR ("fog computing" OR "edge-fog-cloud continuum" OR "vehicular fog computing") AND ("task scheduling" OR "resource allocation" OR "energy efficiency" OR "research gaps") AND ("artificial intelligence" OR "privacy-preserving" OR "multi-objective optimisation") AND ("survey" OR "review" OR "systematic review" OR "tutorial" OR "mapping study").

Following initial search across six databases (IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, Wiley Online Library, and arXiv), 2,627 records were retrieved. Deduplication using Mendeley Reference Manager identified 312 duplicate records based on DOI matching and title similarity, resulting in 2,315 unique records for screening.

The screening process consisted of four stages: (1) Title Screening excluded 1,573 records not explicitly addressing FC, retaining 742 records; (2) Abstract Screening excluded 288 records lacking substantive fog contributions, retaining 454 records; (3) Introduction and Conclusion Review excluded 244 records with insufficient novelty or limitation discussions, retaining 210 records; (4) Full-text eligibility assessment using a structured rubric (minimum threshold: 5/10 points) excluded 154 records, resulting in 56 high-quality papers for final synthesis. Records appearing in multiple databases were retained once, with IEEE Xplore given citation priority.

4.3.2 Search Phase

A keyword-driven search strategy was used to retrieve relevant literature (2015–2025). Figure 4 below illustrates this search strategy. As mentioned, the search strategy focuses on topics related to task scheduling, resource allocation, energy efficiency, AI integration, and privacy.

The primary keywords (PK) of most interest for identifying FC research were FC, AI and Green computing. The secondary keywords (SK) included Edge computing,IoT,

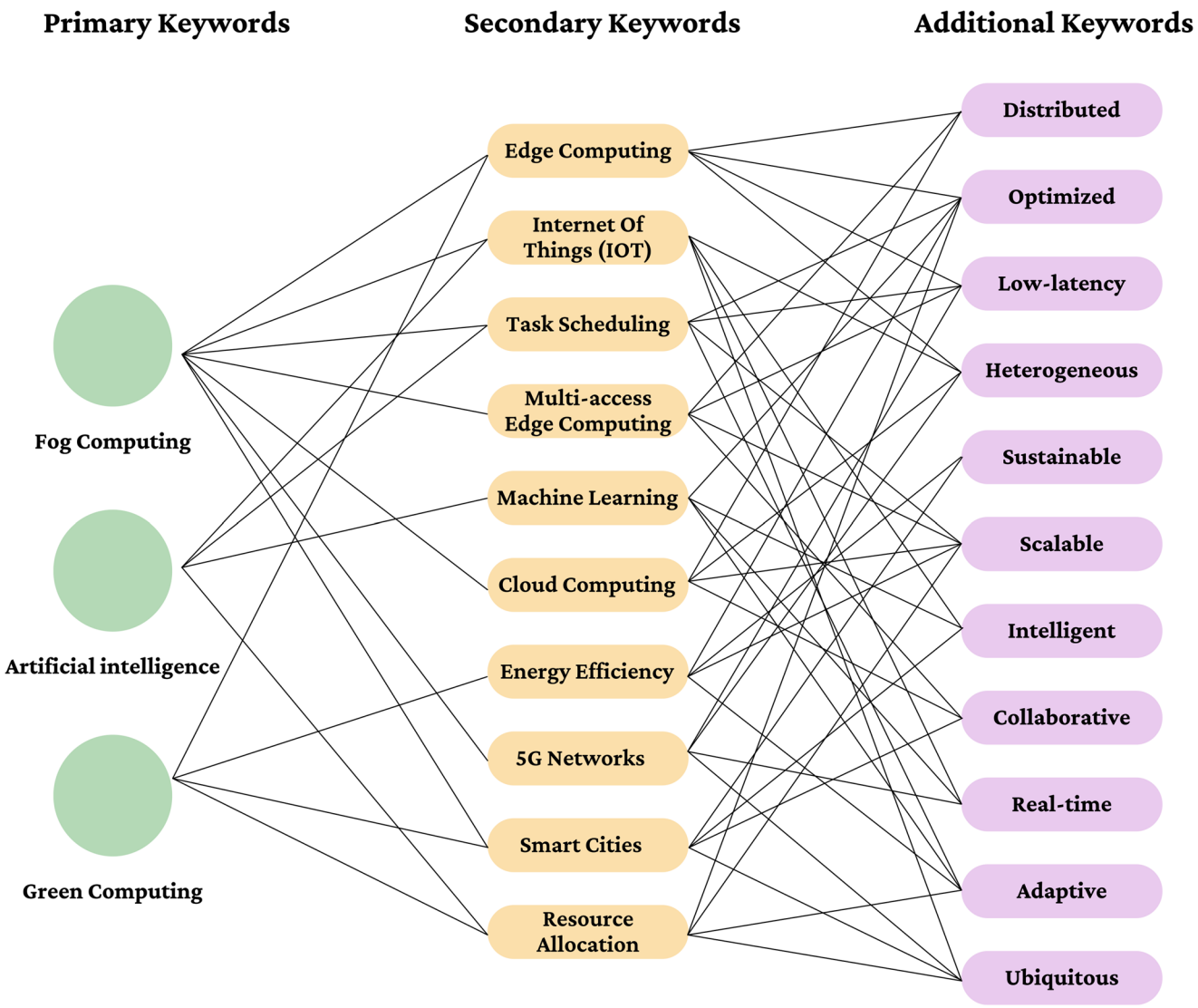


Fig. 4 Mapping of primary, secondary, and additional keywords in fog computing research domain

Task scheduling, MEC, ML, CC, Energy efficiency, 5 G networks, Smart cities, and Resource allocation. The additional keywords (AK) included the terms: Adaptive, Scalable, Sustainable, Hybrid, and Dynamic, and the final search utilised all three categories of keywords, namely primary (PK), secondary (SK), and additional (AK). Combinations including P_k with S_k, followed by A_k, were tested, resulting in all possible pairings: $\forall P_k \wedge \forall S_k \wedge \forall A_k$.

4.3.3 Screening Phase

As a systematic review process, we utilised the component of the knowledge base developed from the variables and conditions included in the original studies (2015 to 2025), and refined it through a three-stage screening process, in an iterative approach, searching the relevant literature based on the abstract, introduction, and conclusion of the papers. Studies were evaluated based solely on predefined inclusion and exclusion criteria, we concealed any disclosure statements associated with the author and publication venue, evaluating the papers purely based on the factual details and conclusions provided in our metadata and the known explanations given. All included studies were logically examined by reviewing the content, as well as the title and abstract, to allow for a balanced assessment in any subsequent conclusions and to mitigate biases indicated by the respective publication venues. Our inclusion criteria included the following variables: IC1 – peer-reviewed studies within FC, IC2 – studies published between 2015 to 2025, and IC3 – studies that mention identified gaps in research or note future work. Our exclusion criteria included: EC1 – studies that do not align with the definition of FC provided above, EC2 – any studies published before 2015, EC3 – studies

not written in English, and EC4 – duplicates from multiple databases.

4.3.4 Inspection Phase

Of the 56 selected studies, the distribution by study type is: simulation-based (33 studies, 59.1%), mathematical modelling (14 studies, 25.0%), and ML/AI-based empirical studies (9 studies, 16%). Additionally, 4 real-world industrial deployments are directly cited as primary evidence sources: SpeziLLM, Cisco IOx, VeReMi vehicular network testbed, and the ENGAGE-HF cardiac monitoring deployment Dataset, Table 10. This distribution reflects the field's current state where simulation dominates evaluation methodology itself identified as a key research gap motivating FogAI-Optimizer's dataset validation layer (Sect. 8, Layer 6)

5 Review Process Reporting

As the SLR process of the randomized sample of studies progressed across six major academic databases, encompassing a comprehensive study of the literature in the included research papers, it is noteworthy to observe the diverse scope of FC research [81, 82]. This review follows a systematic methodology to share a quantitative summary of the discovery, screening, and inspection at each stage as it progresses from keyword search to study selection [83–85]. This report then draws heavily on the knowledge gleaned from existing studies, including research questions about FC paradigms in these investigations [86–88].

Figure 5 presents the study selection process based on inclusion and exclusion criteria. Included studies focus on FC applications, scheduling/resource optimisation, AI/ML techniques, energy efficiency, and MEC, published in peer-reviewed venues (2015 to 2025). Excluded works were non-full papers, not directly related to FC, non-English, cloud-only studies, duplicates, preprints, or superficial contributions. This process ensures only relevant and high-quality studies are analyzed.

5.1 Final Numerical Outcome

Table 3 presents the selection and screening results across major digital libraries. The initial keyword-based search retrieved 2,627 articles. Title screening reduced this to 742 studies, followed by abstract screening, which further narrowed the count to 454 papers. Screening of introduction and conclusion sections resulted in 210 articles. Finally, Full-text eligibility assessment retained 56 high quality studies for detailed analysis. This multi-stage filtering ensured systematic elimination while maintaining research relevance and quality.

Figure 6 illustrates a five-stage systematic filtering process that narrowed 2,627 initial papers down to 56 final selections, representing a 97.9% reduction with a selection rate of 2.1%. studies were sourced from six databases and progressively filtered through title screening, abstract screening, introduction and conclusion review, and full inspection. Of the 56 selected studies, ScienceDirect contributed the highest number with 27 papers (48.2%), followed by IEEE Xplore with 16 papers (28.6%) and Springer with 9 papers (16.1%), while Wiley contributed 3 papers (5.4%) and ACM yielded only 1 paper (1.8%) despite having 393 initial papers, due to its broader computing focus being less aligned with FC-specific research. ArXiv contributed 16 papers at the initial search stage but was entirely excluded at the Quality assessment stage as the review strictly considered only peer-reviewed publications.

5.2 Final Selected Studies

The final cohort of 56 studies delineated in this paper comprises a total of 2.1% of the original studies that met the search criteria. The studies collectively represent those published between the years from 2015 to 2025 and therefore incorporate the legacy of foundational studies, but also new ideas and new contributions to the field. Many studies contributed substantially to the understanding of FC. The studies exhibit considerable methodological diversity. Simulation-based studies accounted for the most in the selection of 33 studies (59.1%), which utilised specific frameworks (e.g., iFogSim, CloudSim) or their own simulations to examine FC architecture or algorithms. Studies utilising mathematical modelling methods, or approached via mathematical performance evaluation approaches, accounted for 14 studies (25.0%), which investigated applications of optimisation or game theory or evaluated performance via mathematical modelling. Finally, studies utilising ML methods accounted for nine studies (16.1%) that employed various types of neural networks, RL, and clustering in intelligent FC systems. The studies reflected a significant coherence with the defined scope of the research topic focus. The largest category was task scheduling/resource management with 21 studies (37.5%). These studies examined challenges and research issues critically related to FC environments. The next largest category was AI with 16 studies (28.6%). This category of studies reflects a growing interest in intelligent fog master work environments for task allocation. The third largest category was vehicular FC with 12 studies (21.4%), which represents the most relevance to transport and mobility via FC environments. The smallest category was green computing and energy efficiency with seven studies (12.5%),

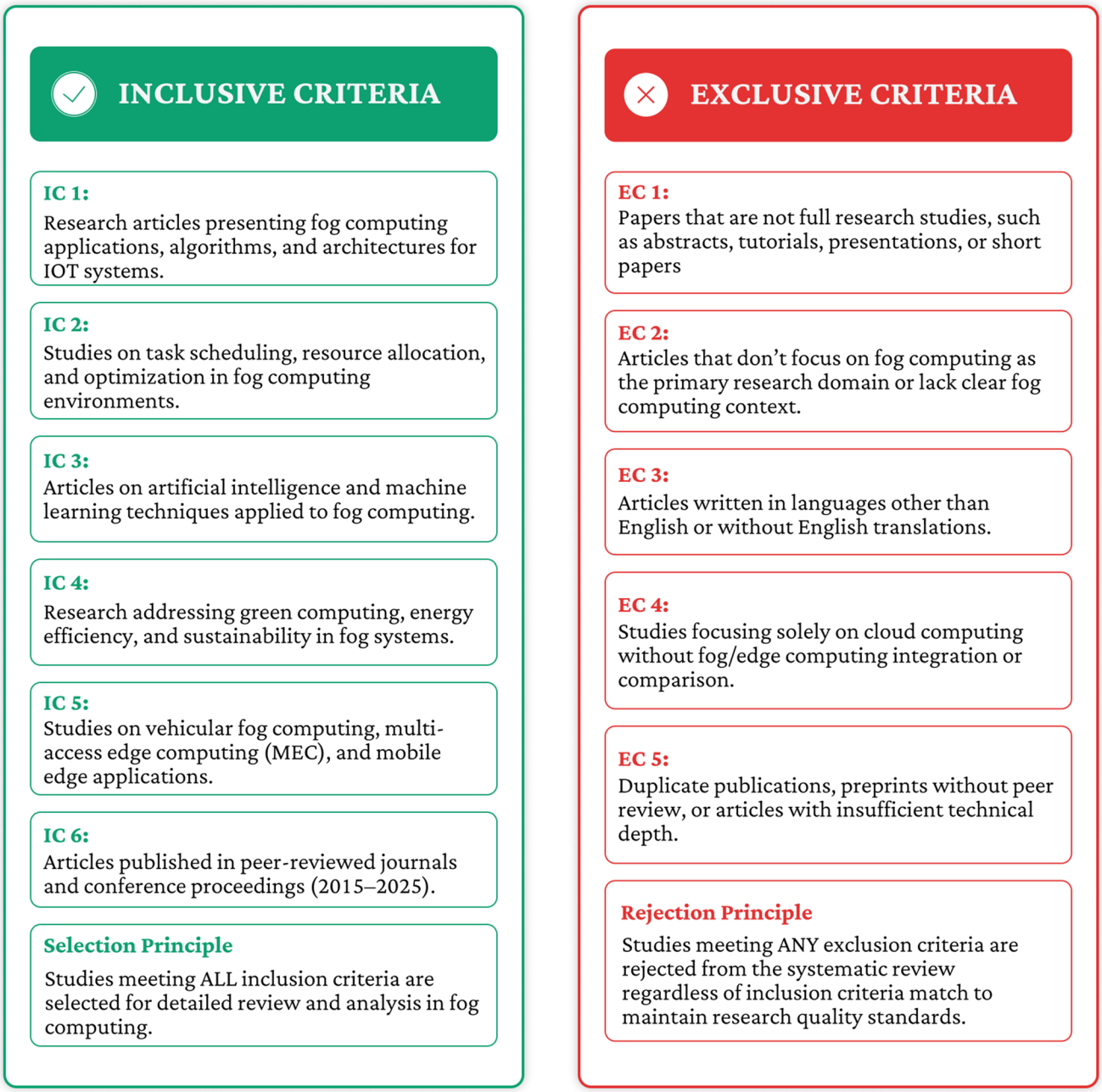


Fig. 5 Criteria for study selection based on inclusion criteria and exclusion criteria

Table 3 Selection and screening results from digital libraries

Selection Criteria	ACM	Archive	IEEE	ScienceDirect	Springer	Wiley	Total
Search based on Keywords	393	16	945	468	763	42	2627
Screening based on Title	40	3	454	135	89	21	742
Screening based on Abstract	8	0	263	96	79	8	454
Screening based on Introduction and Conclusion	1	0	84	64	57	4	210
Inspection based on Full Article	1	0	16	27	9	3	**56**

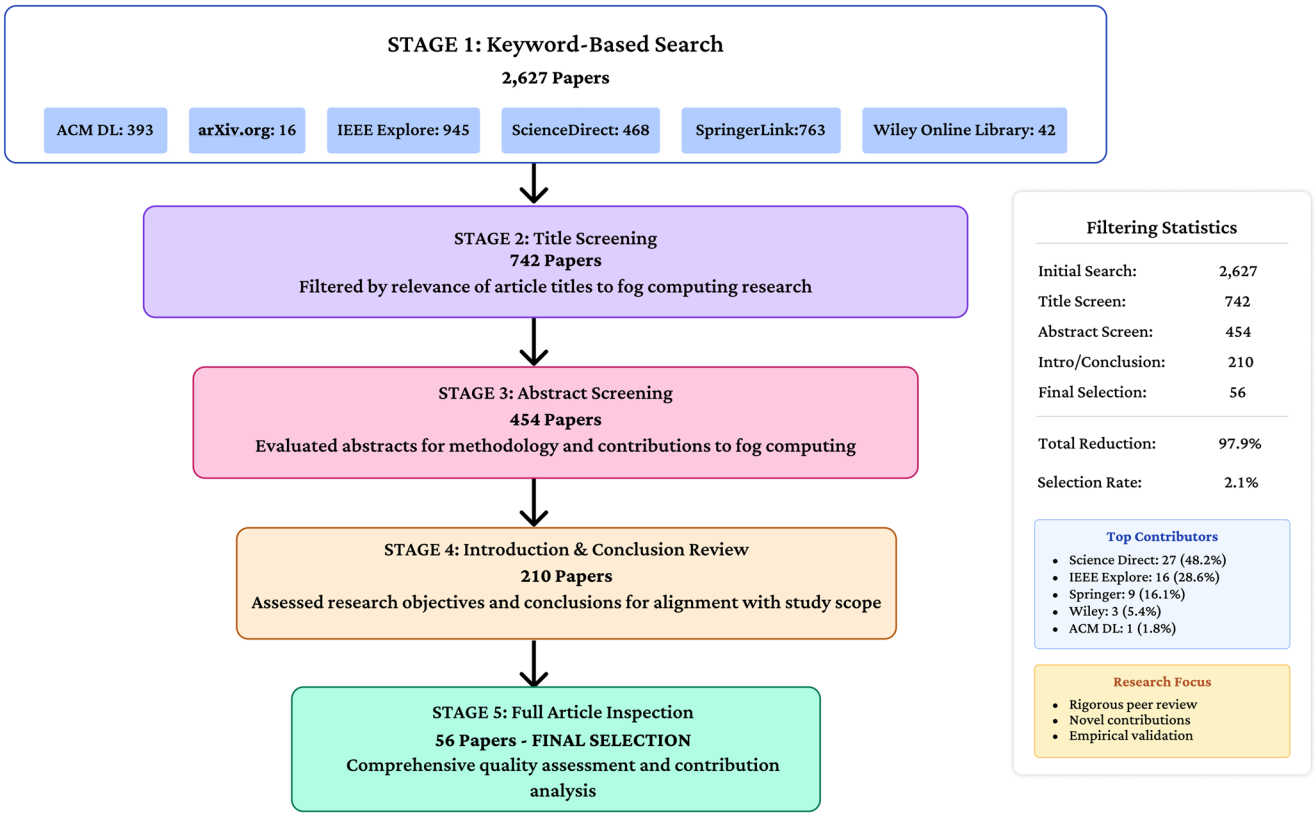


Fig. 6 Multi-stage filtering process

typical for the growing concern for sustainability trends in FC research. The geographic distribution of research contributions showed a predominance of studies coming from the different geographic areas, with Asia-Pacific institutions represented (27 studies, 48.2%) institutions accounting for studies primarily from China, India, and Australia. The second geographic distribution was institutions contributing studies from Europe (17 studies, 30.4%) from Germany, the United Kingdom, Italy, and the Netherlands. North American studies were conducted in institutions from Canada and the United States, with a total of 12 studies (21.4%).

The publication venues of studies reiterated all findings about institutional contributions. There tends to be a concentration of studies in high-impact journals and conferences. The IEEE Transactions journals contributed 19 studies (33.9%) contributing including IEEE Transactions on Mobile Computing, IEEE Transactions on Vehicular Technology, and IEEE Transactions on CC. The contributions from Elsevier Journals were the largest number of studies published (22 studies) and comprise about 39.3% in Future Generation Computer Systems, Computing Networks, and Journal on Network and Computer Applications. The contribution from Springer was 10 studies, 17.9%. They also recognise the minimum contribution, where ACM, Wiley contributions total five studies, 8.9%. Nonetheless, the temporal distribution indicates more towards academic research being conducted in recent years. The earliest time period of studies published (2019–2021) contributed 16 studies (28.6%), depicted as being foundational architectures and basic Optimisation methodologies. The middle time period of studies published (2022–2023) contributed 25 studies (44.6%), demonstrated here as key contributions integrating AI and advanced scheduling algorithms. The most recent time period (2024 to 2025) contributing to this cohort of studies, 15 studies (28.6%) focused on contemporary and cutting-edge frameworks, including green computing and next-generation networking-based frameworks.

5.3 Quality Assessment of Selected Studies

To differentiate between empirical, simulation-based, and conceptual studies within the final cohort of 56 studies, a three-criterion Quality assessment framework was developed. Each study was scored across three dimensions: (1) Empirical rigor whether the study uses real world deployment (score: 3), simulation with recognized tools such as iFogSim or CloudSim (score: 2), or purely conceptual/mathematical analysis (score: 1); (2) Reproducibility whether the study provides a publicly accessible dataset or code repository (score: 3), partial validation data (score: 2),

or no reproducible artifacts (score: 1); and (3) Novelty of contribution assessed as high (score: 3), medium (score: 2), or low (score: 1) based on whether the proposed method outperforms three or more baselines, two baselines, or fewer respectively. Total scores range from 3 (minimum) to 9 (maximum). Studies scoring 7–9 are classified as high quality, 4–6 as medium quality, and 3 as low quality. Of the 56 selected studies, 9 (16.1%) achieved high quality scores, 33 (58.9%) achieved medium quality, and 14 (25.0%) were conceptual or early stage works rated as foundational. This distribution reflects the field's current bias toward simulation based evaluation, which is itself identified as a key research gap addressed by FogAI Optimizer's dataset integration layer.

This rubric was designed specifically for simulation- and algorithm-based computing research, as standard biomedical Quality assessment instruments are not transferable to this domain. The criteria reflect three widely accepted dimensions of contribution quality in systems research: empirical grounding, reproducibility, and comparative novelty. Scores were assigned by the primary reviewer and verified by the second author for consistency.

6 Findings of the Review

6.1 Drivers of Advanced FC Paradigms

Prediction-based approaches represent a forward-looking direction for latency reduction demonstrate that workload prediction integrated into fog scheduling decisions reduces service delay by 18% to 26% [51] compared to reactive scheduling while maintaining Quality of Service (QoS) guarantees for heterogeneous smart city services. Similarly [14], show context-aware IoT service frameworks reduce service delay by 23% through intelligent edge-fog workload routing. Advanced FC paradigms, as highlighted by several interrelated factors, address and provide solutions to various problems encountered in IoT environments. These motivations are identified from iterations of the referenced studies, as well as from the cited existing resources. The motivation for deploying FC at the edge is to enable instantaneous data processing for IoT applications, including autonomous vehicles and telehealth [89]. The authors of [90], in their work on vehicular FC, have addressed latency awareness in their dispatching task with DRL scheduling and report a reduction in service latency by up to 12% in vehicular FC, emphasizing its value for time-sensitive activities [91]. The rapid increase in IoT data (e.g., 1GB/hour from a smart camera) means that local processing of data is essential to alleviate the strain on bandwidth [11]. Other researchers have suggested that FC can reduce cloud transmission data by up to 40%, not only lowering operational costs but also mitigating network congestion. IoT is becoming increasingly distributed and is therefore susceptible to compromised data, privacy breaches, and exposure [92]. Accordingly, [93] comments that local, real-time processing of sensitive data is one of the key characteristics of FC and cites edge-level encryption, which can help mitigate privacy issues in IoT environments, as a motivating factor for the fog approach to improving security for IoT environments [94]. As IoT continues to evolve, and energy is a growing concern in the advanced computing field, green fog paradigms arise from the motivation to create frameworks that can integrate renewable energy sources into IoT systems [67], and [95] considers the use of ongoing climate change environmental variables that integrate sustainable fog architectures, or other concepts like [96], which conceptualizes AI-related IoT-based air quality calculations that contribute to reducing the carbon footprint. Another reason for advanced fog paradigms stems from the mobility of IoT devices, similarly those in a car, and motivates, for example, adaptive fog systems [95]. For instance, they address system mobility with mobility-aware task scheduling mechanisms, which facilitate application operation and movement from one network to another in a non-disruptive manner.

Finally, since IoT places demands on the network environment in terms of scalability, fog distributed architecture solutions offer several potential solutions [97]. For example, [98], their work in vehicular networks increases throughput and suggests scalability issues in resource orchestration that are also motivating [89]. The relevance of AI is extremely prominent, especially at the edge, like in DDPG edge allocation, which is a motivating factor for an advanced paradigm around the edge, guided by these examples, integrating advanced paradigms into large and smart application literature or fields similarly urban traffic management [45].

Figure 7 illustrates a structured flow diagram that maps seven core IoT challenges to their respective FC solutions, all ultimately converging into a single unified outcome. The diagram is color-coded into three categories red for IoT challenges, green for fog solutions, and blue for final integration. IoT challenges include data overload, delays, network waste, security risks, power drain, mobility issues, and scalability limitations, each of which is directly addressed by a corresponding fog solution such as local data filtering, real-time edge computation, bandwidth optimisation, local privacy measures, green energy efficiency, adaptive mobility support, and AI-powered scalable architectures respectively. All these interconnected fog solutions collectively converge into a unified outcome of distributed, edge-smart designs optimized for next-generation 6G IoT networks, demonstrating that FC serves as a comprehensive

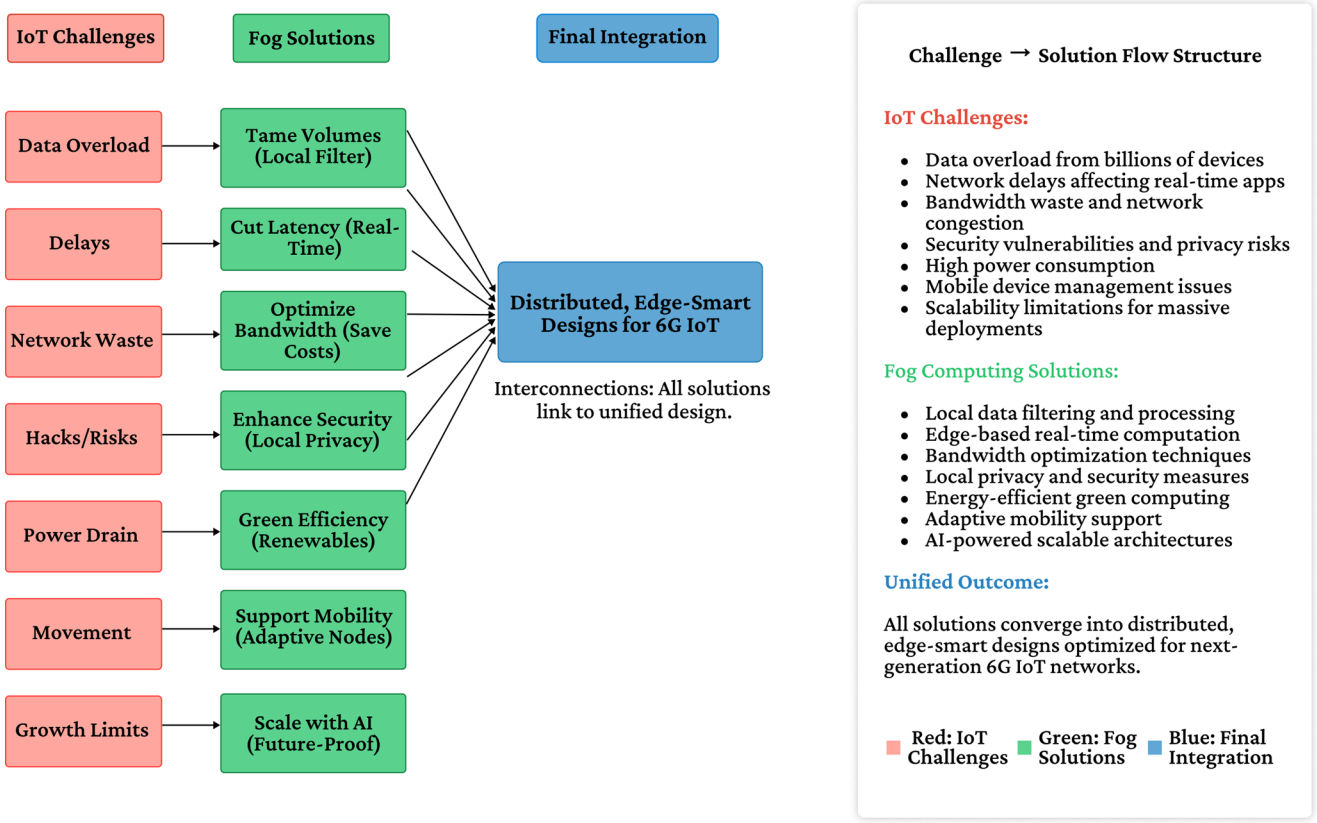


Fig. 7 IoT challenges and their corresponding fog computing solutions

and future-ready solution to the growing challenges of IoT ecosystems.

Table 4 summarizes the driving forces behind advanced FC models in IoT contexts. It contains four columns: Motivational Drivers (latency reduction, etc.); IoT Challenge (high latency of 100–500 ms); Evidence from Research (12% reduction in latency; and Innovation Opportunities (real-time 6 G applications).

6.2 Challenges Addressed by FC

The distributed nature of fog processing inherently provides redundancy with data processing across the distributed fog rather than relying on data processing at a single point within the cloud. Collectively, these modifications ensure that resource engagement in IoT environments is optimal, which is supported by various approaches under review in previous studies. The issues addressed by the FC approach are a broad spectrum of issues, including latency, bandwidth, energy, security, mobility, resource heterogeneity, and reliability [60]. The evidence collectively presented in this SLR, along with the peer contributions of the reviewed studies and the common knowledge of the cited articles, provides a common ground of knowledge regarding better mitigation for these associations. From this, there are significant instances where innovation can be appropriately future-directed in FC systems, particularly in dynamic environments like smart cities or vehicular networks.

Figure 8 categorizes the FC challenges across three overlapping themes: Performance, Security, and Resource Management, converging into a unified outcome of enhanced IoT efficiency.

Table 5 summarizes how FC methods address eight key IoT challenges aligned with RQ5, including Latency Reduction, Bandwidth optimisation, Energy Efficiency, Scalability, Security and Privacy, Mobility, Resource Heterogeneity, and Fault Tolerance. Each entry identifies the impacted IoT aspect, the mitigation approach used, the quantified benefit cited from respective papers, and a research insight suggesting future directions for that area.

Table 6 maps the quantitative performance claims of this survey to their primary source studies. Latency reductions of 50% to 70% are supported by simulation-based studies in healthcare and smart city fog deployments, while real-world validation via SpeziLLM confirms 40% to 60% latency gains. Bandwidth savings of up to 80% are evidenced through Cisco IOx industrial deployments, and AI-driven efficiency improvements of 20% to 40% are reported in RL-based scheduling studies. Together, these results span

Table 4 IoT-Powered fog motivation matrix

Ref.	Motivational Driver	IoT Challenge	Evidence from Research	Innovation Opportunity
[90]	Latency Reduction	High end-to-end delay (100–500 ms)	Vehicular FC reduces latency by 12%	Real-time 6 G-enabled IoT applications
[99]	Bandwidth Optimisation	High data transmission load (≈ 1 GB/hour per device)	Fog preprocessing reduces cloud-bound data by 40%	Cost-efficient and scalable IoT networks
[93]	Privacy and Security	Vulnerability to data breaches	Fog-layer local encryption improves data privacy	Quantum-resistant and zero-trust architectures
[96]	Energy Efficiency	High energy demand in IoT deployments	Green IoT approaches achieve measurable carbon reduction	Integration of renewable and energy-aware IoT systems
[100]	Mobility Support	Frequent disconnections due to device movement	Adaptive fog scheduling improves mobility support	Seamless 5 G/6 G edge handoff mechanisms
[98]	Scalability	Rapid growth of IoT devices and network traffic	Fog/edge deployment improves throughput and load distribution	Dynamic and scalable resource orchestration
[101]	Edge Intelligence	Limited on-device computation	DDPG-based resource allocation enhances decision quality	AI-driven smart city cognitive infrastructures

simulation, industrial, and real-world environments, grounding the abstract claims in verifiable literature evidence.

6.3 Ai/ML-Based Privacy-Preserving and Energy-Efficient Methods

FC acts as a crucial intermediary layer between IoT devices and the cloud, enabling local processing of IoT data to reduce latency in scenarios like smart city surveillance for traffic management. However, operating in a rapidly dynamic environment where numerous uncertainties arise poses a risk of potential privacy breaches and energy inefficiencies. Therefore, AI and ML can provide timely perspectives on managing privacy protection and energy conservation, as both domains rely on intelligent decision-making to mitigate information risks as data is being sensed and processed by computing capabilities [23]. For example, in an urban ecosystem, as discussed in [113], a fog architecture needs to effectively and securely accommodate or handle urban IoT data streams, where AI/ML approaches help preserve individuals' personal information from IoT devices while curbing energy consumption, since edge nodes operate on batteries with limited resources. Thus, the question considers the approaches available in Fig. 9, which are identified as privacy-oriented, energy-oriented, or hybrid, with justification from recent literature references in considering the mechanisms, applications, and limitations [40].

Privacy-preserving AI/ML approaches focus on protecting sensitive data during processing for fog systems. FL with differential privacy allows sharing only aggregate updates and training on models at the local level by adding noise to individual user data so that it is not reconstructible. FL with differential privacy can be useful in dynamic smart cities that collaboratively analyze traffic patterns without exposing user locations [107, 114]. Homomorphic encryption can perform ML operations on encrypted data, where fog nodes perform computations without first decrypting the data. This encryption technique is suitable for urban health monitoring as it will maintain the privacy of individual medical data during real-time analysis [111, 115]. AI-based anomaly detection uses ML to identify unusual or suspicious patterns, like potential hacks, while operating on datasets that are "masked." For example, fog systems in cities can identify security threats in real time without having access to all the data. Energy-efficient AI/ML methods strive to minimize power consumption while maintaining performance under varying conditions. DRL, a form of advanced AI/ML technique, can learn through trial and error what the optimal resource allocation policies are. This mechanism has been employed in smart city sensor networks to reduce energy usage during off-peak times, allowing fog nodes to dynamically offload tasks, and allowing processors or processing

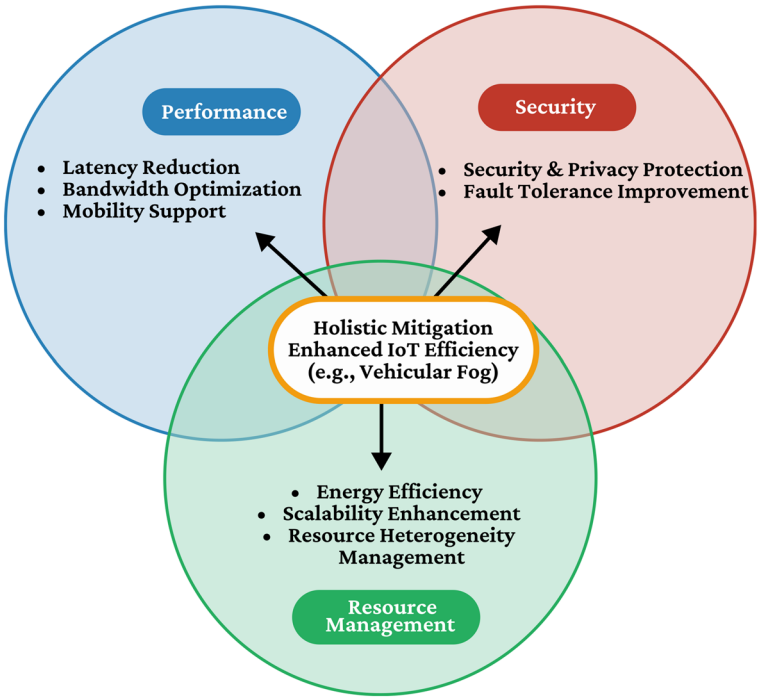


Fig. 8 Challenges mitigated by fog computing

Table 5 Mitigation impact summary across IoT challenges

Ref.	Challenge	Impacted IoT Aspect	Mitigation Approach	Quantified Benefit	Research Insight
[102]	Latency Reduction	Real-time responsiveness	Edge-based low-latency processing	12% latency reduction	Enhances 5 G/6 G real-time services
[103, 104]	Bandwidth Optimisation	Network traffic load	Fog-layer data filtering and preprocessing	40% reduction in cloud-bound data	Improves bandwidth scalability
[17, 105]	Energy Efficiency	Device and node power usage	Localized computation and adaptive duty cycling	Lower energy consumption	Supports renewable energy-aware IoT
[106, 107]	Scalability Enhancement	Handling massive device growth	Distributed workload balancing	Higher throughput under load	Enables large-scale IoT deployments
[106, 108]	Security and Privacy	Data confidentiality and integrity	Fog-level encryption and lightweight security	Improved privacy protection	Advances in zero-trust, post-quantum IoT
[109, 110]	Mobility Support	Connectivity resilience	Mobility-aware fog node handoff	Seamless device transitions	Boosts 5 G/6 G mobility performance
[103, 111]	Resource Heterogeneity	Diverse hardware and capabilities	Intelligent resource orchestration	Balanced, efficient resource utilization	Moves toward unified fog–edge frameworks
[100, 112]	Fault Tolerance	Service reliability	Redundant edge nodes and failure-aware scheduling	Reduced service downtime	Strengthens fault-tolerant IoT architectures

Table 6 Key performance claims and sources

Ref.	Metric	Reported Gain	Environment	Validation Method
[33] Hazra et al.	Latency	50% to 70% reduction	Smart healthcare and vehicular fog vs cloud	Simulation (iFogSim)
[20] SpeziLLM	Latency	40% to 60% reduction	Healthcare fog vs cloud	Real-world iOS app
[21] Cisco IOx	Bandwidth	Up to 80% savings	Edge preprocessing	Industrial deployment
[40] Ghafari et al.	System efficiency	20% to 40% improvement	iFogSim simulation	Simulation
[36] Songhorabadi et al.	Latency	50% to 70% reduction	Smart city fog	Simulation

components to go to sleep or power down completely. Another energy-efficient AI/ML method is based on the use of evolutionary algorithms, along with genetic or particle swarm optimisation techniques [55]. Similar to DRL, evolutionary algorithms evolve a solution at each iteration to find the configuration with the minimum energy. This method is particularly useful in any urban FC and EC setup where resources and workloads need to be balanced across heterogeneous nodes [111, 116].

There are hybrid approaches that combine/utilise both privacy considerations and energy efficiency benefits, as applicable. AI-enhanced edge analytics processes data that

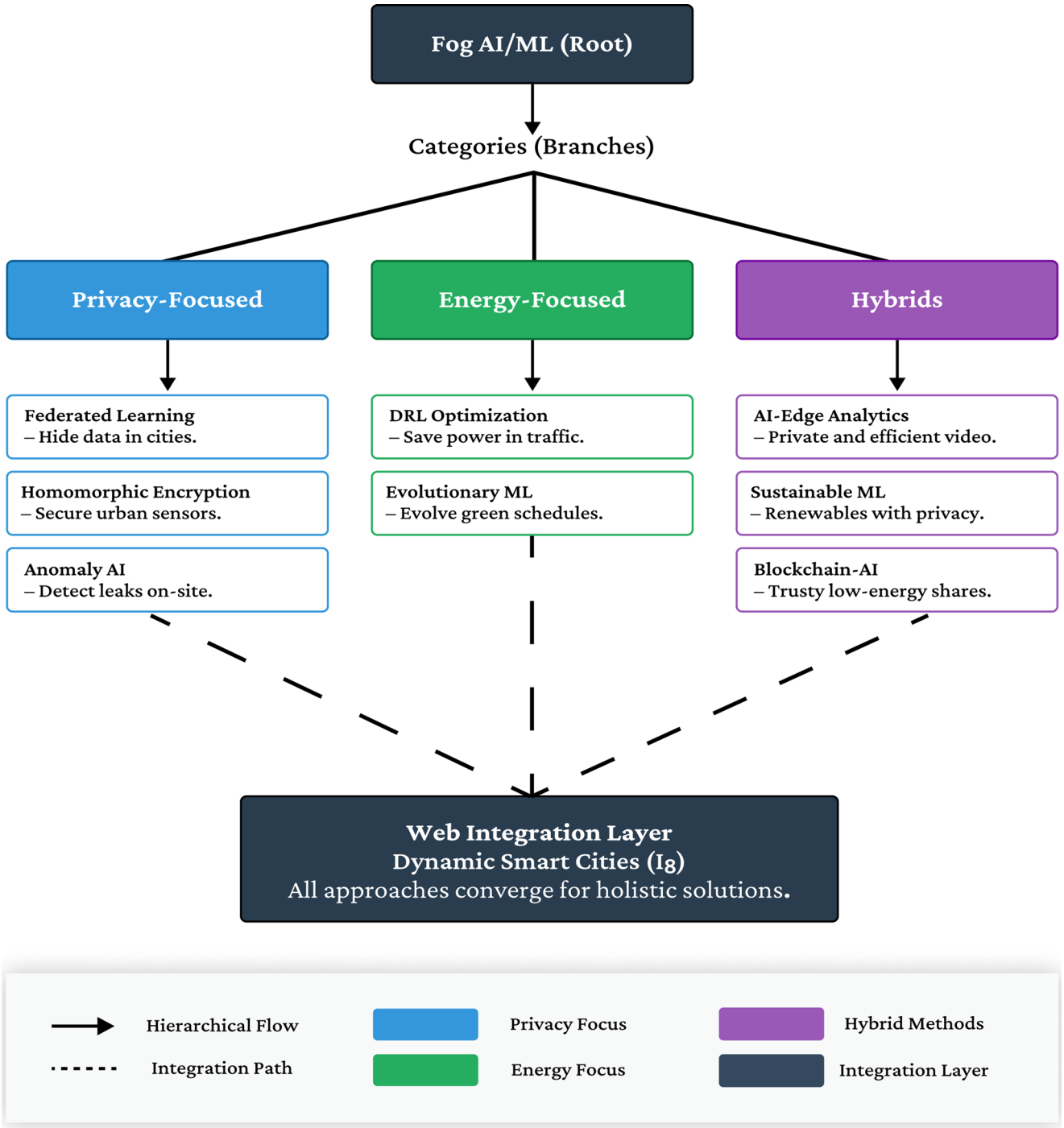


Fig. 9 Taxonomy of fog AI/ML approaches for dynamic smart city applications

is potentially sensitive at the fog layer at the network edge, while also employing encryption methods to incorporate low-power computation actions, for example, with city fog systems for video surveillance, where data security and longevity are crucial, applicable to real-time video surveillance applications at scale [107, 117]. Sustainable ML, categorized by its focus on privacy communities and frameworks, and blockchain, combines renewable energy management with secure elements of the infrastructure, enabling the distribution of green energy in urban grid systems while maintaining user anonymity [107, 112]. Multi-modal ML fuses two or more data types (audio and video), and this combination generates more information to enable better decision-making and pattern development in both efficient and privacy-preserving functions for city infrastructure fog systems, while reflecting energy constraints for a nuanced

and multi-layered approach to privacy constraint removal. These methods appear promising, but often require more collaboration to see proper use cases and real-world deployments in real-world 6 G scenarios [114, 118].

The relationship between these technologies, both individually and collectively, highlights their potential for synergy. For example, privacy-preserving methods like FL reduce energy consumption in data transmission, while hybrid approaches can create employment opportunities in smart city applications with various constraints [113].

Figure 10 shows the AI/ML data processing flow in fog environments. IoT data first undergoes a privacy check if required, privacy-preserving methods such as FL are applied locally; otherwise data is redirected to a secure node. Data is then processed at the fog layer with encryption. If energy optimisation is needed, DRL-based resource allocation is applied; otherwise standard monitoring continues. When both privacy and energy constraints exist simultaneously, hybrid approaches such as sustainable ML with blockchain handle both requirements, producing secure, energy-efficient output for real-time applications.

Table 7 presents six AI/ML approaches for privacy preserving and energy efficient FC in smart city environments. Privacy centric methods FL, homomorphic encryption, and anomaly detection protect sensitive data during local processing. Energy-centric approaches DRL and evolutionary algorithms optimize resource allocation under dynamic loads. Hybrid methods combine both goals, supporting applications ranging from urban surveillance and

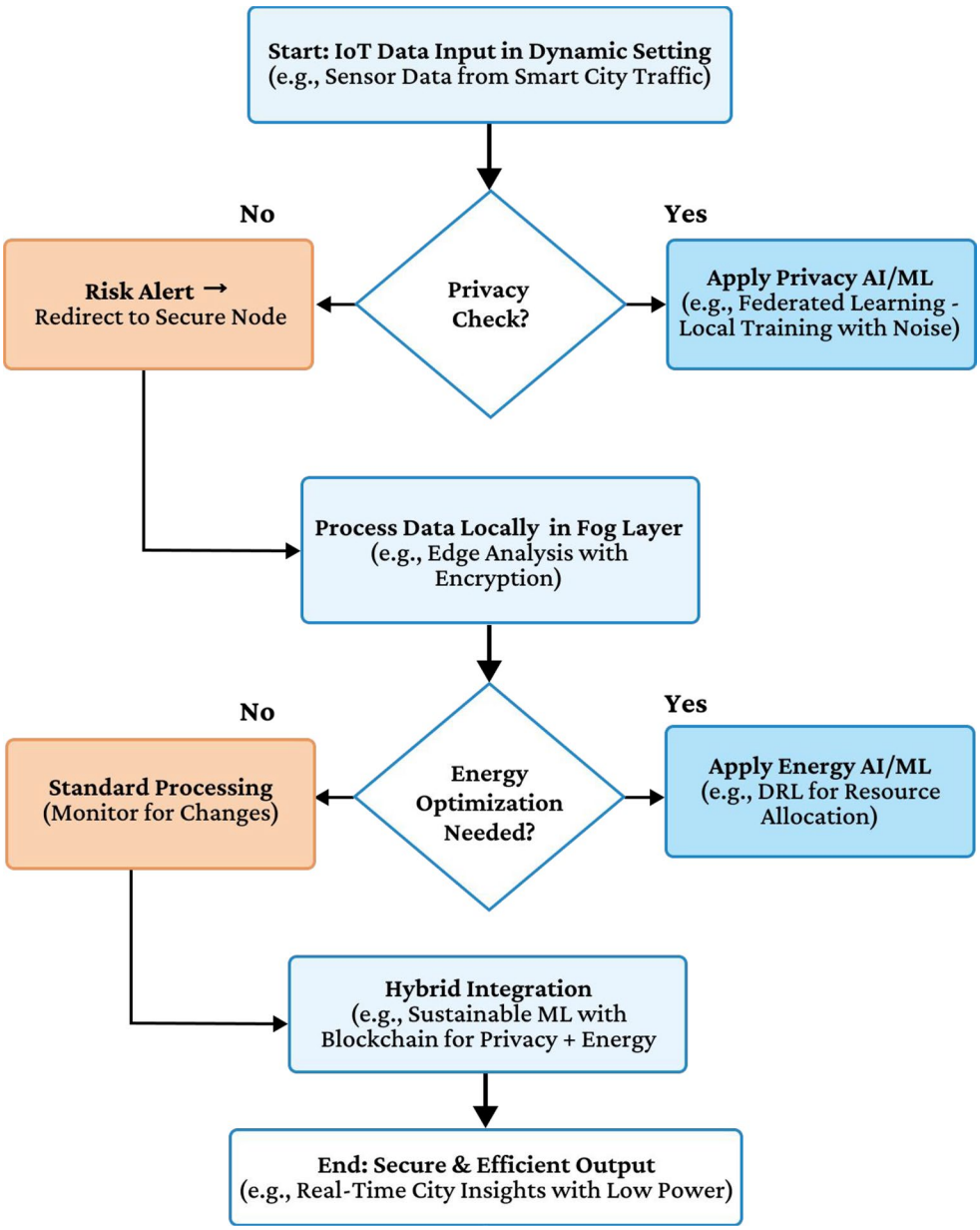


Fig. 10 Secure and efficient IOT data processing flow

Table 7 Smart city approaches

Ref.	Approach	Privacy Mechanism	Energy Strategy	Dynamic Adaptability	Smart City Use Case
[112]	Proactive Edge Surveillance	On-device video/sensor processing prevents external exposure	Saves energy by reducing continuous cloud uploads	Rapid response to real-time security events	Urban security monitoring
[113]	Distributed Data Processing	Local preprocessing reduces data exposure during transmission	Minimizes energy by processing at the network edge	Adapts to changing traffic and mobility patterns	Traffic congestion and pattern analysis
[119]	ML-Based Resource Optimisation	Secure processing on optimized fog/edge nodes	AI-driven demand prediction reduces unnecessary power usage	Adapts to heterogeneous IoT device loads	Smart transportation and fleet management
[113]	Application-Aware AI Allocation	Localized data handling based on app sensitivity	Allocates energy based on application priority	Adjusts resources to app-specific workloads	Urban healthcare and remote patient monitoring
[112]	DRL-Based Resource Management	Not applicable (focus is energy Optimisation, not data privacy)	DRL optimizes resource allocation under fluctuating demand	Responds to variable network and sensor load	Adaptive sensor scheduling in smart environments
[119]	ML-Assisted Fog Offloading	Minimizes cloud exposure by retaining data at fog nodes	Reduces network energy usage by lowering cloud communication	Adapts to dynamic interactions among IoT devices	Smart city data aggregation and offloading

traffic management to healthcare monitoring and smart transportation.

6.4 Architectural Mechanisms in Fog Systems

FC architecture is a framework that organizes the sharing of data processing and management tasks between IoT devices and cloud environments to facilitate local execution and address limitations imposed by latency in the cloud ecosystem. FC facilitates localized data processing and connectivity while allocating computational resources closer to the location streaming the data sources and disrupting traditional CC by establishing a dedicated layer. The fog architecture comprises a framework that includes a distributed layer of devices providing a middle layer of computing and processing speed and reliability for dynamic IoT applications. Architectural mechanisms form the basis of the FC architecture, similarly layered structures to create a controlled way of managing data flow, node management capabilities for resource distribution, and methods for integrating connectivity. The architectural mechanisms are not fixed but adaptive. In environments similarly mobile vehicular networks, fog systems require adapting the design by monitoring filter applications for varying environmental patterns. Indeed, a careful reading of the existing literature reveals that the architectural mechanisms primarily focus on the concepts of scalability and security; various mechanisms are defined, similarly the layered structure which they defined and explored based on six layers, including a device layer at the bottom, a fog server layer in the middle, and a cloud local for storage at the top. They also elaborate on the industrial context of edge-fog integration discussed in [120].

Figure 11 illustrates the three categories of FC architectural techniques. Layered structures hierarchical and decentralized organize data flow from edge devices through fog nodes to the cloud, as seen in smart city traffic and factory monitoring scenarios. Node and resource management techniques, including resource orchestration and node federation, enable dynamic allocation and collaborative learning for vehicular and urban applications. Integration methods connect the fog layer with cloud infrastructure through hybrid approaches, supporting use cases such as urban healthcare analytics. Together these techniques form an adaptive architectural foundation for dynamic IoT environments targeting 6 G readiness.

Table 8 summarizes five key FC architectural techniques across four dimensions: structural design, operational role, deployment scenario, and enhancement potential. Hierarchical fog layering manages tiered data flow for smart city traffic, while decentralized edge layering enables autonomous processing for industrial monitoring. Resource orchestration, node federation, and hybrid cloud-fog integration collectively address vehicular load balancing, secure urban analytics, and healthcare response systems respectively, each with identified optimisation opportunities.

6.5 Task Scheduling and Resource Management

FC offers alternatives for task scheduling and resource management in the fog ecosystem. It enables data processing close to IoT devices rather than solely in remote cloud systems, which is crucial for latency sensitive applications. Scheduling tasks primarily involve allocating different tasks for sensor data analytics in a sparse ubiquitous environment with fog nodes distributed across the entire region, using limited resources from battery powered IoT devices and dynamic cloud context. These distributed, resource constrained characteristics present challenges due to lack of centralized control. Intelligent AI-based solutions are

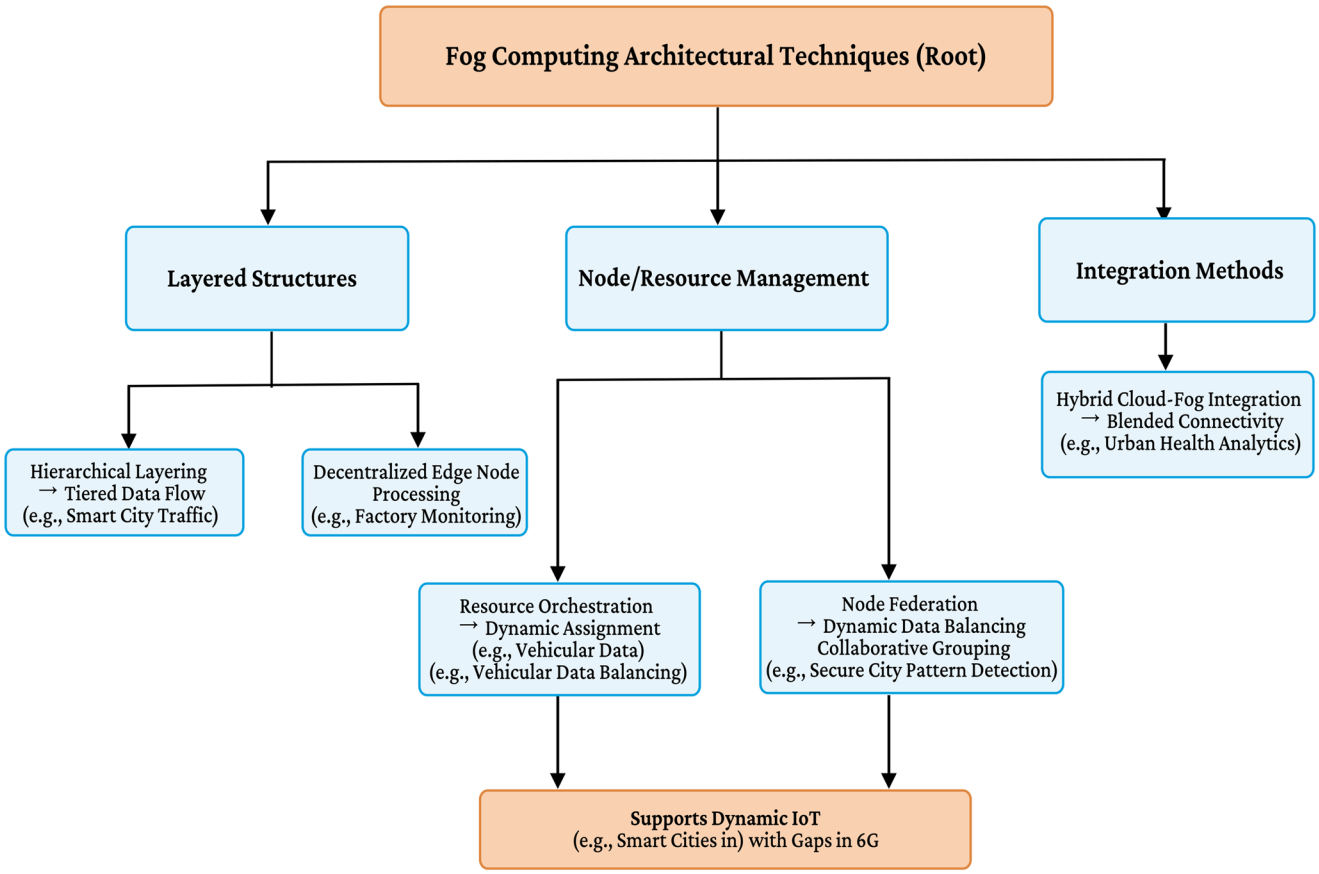


Fig. 11 Fog computing architecture techniques

Table 8 Overview of Fog/Edge architectural techniques and their characteristics

Ref.	Technique Name	Structural Design	Operational Role	Deployment Scenario	Enhancement Potential
[121]	Hierarchical Fog Layering	Multi-tier fog–edge–cloud architecture	Routes and prioritizes data across tiers	Smart city traffic flow management	Improve inter-tier coordination and latency control
[120]	Decentralized Edge Layering	Distributed peer-to-peer edge nodes	Enables autonomous local decision-making	Industrial vibration and anomaly monitoring	Enhance node trust, security, and robustness
[122]	Resource Orchestration Framework	Dynamic resource allocation grid	Distributes CPU, bandwidth, and power adaptively	Vehicular fog data load balancing	Reduce orchestration overhead and optimize fairness
[123]	Node Federation Model	Federated cluster of cooperative nodes	Supports collaborative learning and shared inference	Secure urban pattern detection and analytics	Minimize synchronization delay and improve model sharing
[124]	Hybrid Cloud–Fog Integration	Combined cloud–fog bridge architecture	Balances latency-sensitive local tasks with cloud heavy-compute tasks	Urban healthcare and emergency response systems	Improve coordination efficiency and data consistency

needed to consider multiple objectives such as time and energy [44]. Based on this analysis of studies, some of these topics have been validated as essential aspects for the concrete IoT domain, similarly smart cities where waiting times increase congestion. FC environment poses major problems from a task scheduling and resource control perspective for executing effective, timely plans [125]. A paramount factor is latency and duration in a dynamic environment, where although people may be at an arbitrary point, the task will slow down or take considerably longer due to distance or client capacity [51]. From a planning perspective, this creates a significant bottleneck effect in real-time IoT systems like those in ubiquitous computing. The situation is particularly difficult in FC because, by definition, fog, as highlighted in [126], involves dynamic resource availability, where a fog node is never a single node, or the same node supports different resources.

Energy consumption in fog systems rises during peak loads due to layered processing demands, which is particularly challenging in green IoT scenarios where dynamic

workloads must be balanced against renewable energy availability [127]. Task assignment in distributed fog environments introduces inherent security and privacy risks tied directly to the tasks themselves. These risks arise from operational procedures executed across separate, autonomous nodes, where assignments must balance efficiency with isolation to prevent vulnerabilities like data leakage or unauthorised access. A comprehensive review of data protection requirements for fog-cloud operations is provided in [128]. The challenges of mobility and load balancing result in rescheduling when the mobility of devices causes resource imbalances, adding to the balance problem in mobile fog environments noted in [129]. Finally, multi-objective conflicts arise [130] when a single objective goal competes with other objectives in heterogeneous fog environments.

Figure 12 maps six FC task scheduling problems to their corresponding solutions. Latency and mobility are addressed by DRL-based scheduling, resource scarcity by hybrid evolutionary optimisation, energy overload by cuckoo search-based caching, security risks by graph RL with FL, mobility imbalance by DDPG adaptation, and multi-objective conflicts by Harris Hawks optimisation. Solving one problem creates networked benefits across others. Real-world validation and quantum security threats remain the primary identified gaps.

Figure 13 begins with the deployment flow for FC solutions, starting with identifying the core challenge such as latency, resource scarcity, or IoT data issues, and applying corresponding AI-driven solutions like DRL scheduling, Cuckoo Search, and FL. The solution is then implemented with real-time monitoring, and if performance does not improve, it is refined and retested in a continuous loop. Once performance is satisfactory, the flow concludes with enhanced fog efficiency, such as smart city improvements, while noting that real-world validation and quantum threats remain as future open challenges.

Table 9 maps six core FC challenges latency, resource scarcity, energy consumption, security, mobility, and multi objective conflicts to their corresponding AI driven solutions, performance gains, and proposed innovations. Each row identifies a root cause, a solution technique such as DRL-based scheduling and a quantified improvement, providing a structured problem-to-solution reference for FC task scheduling research.

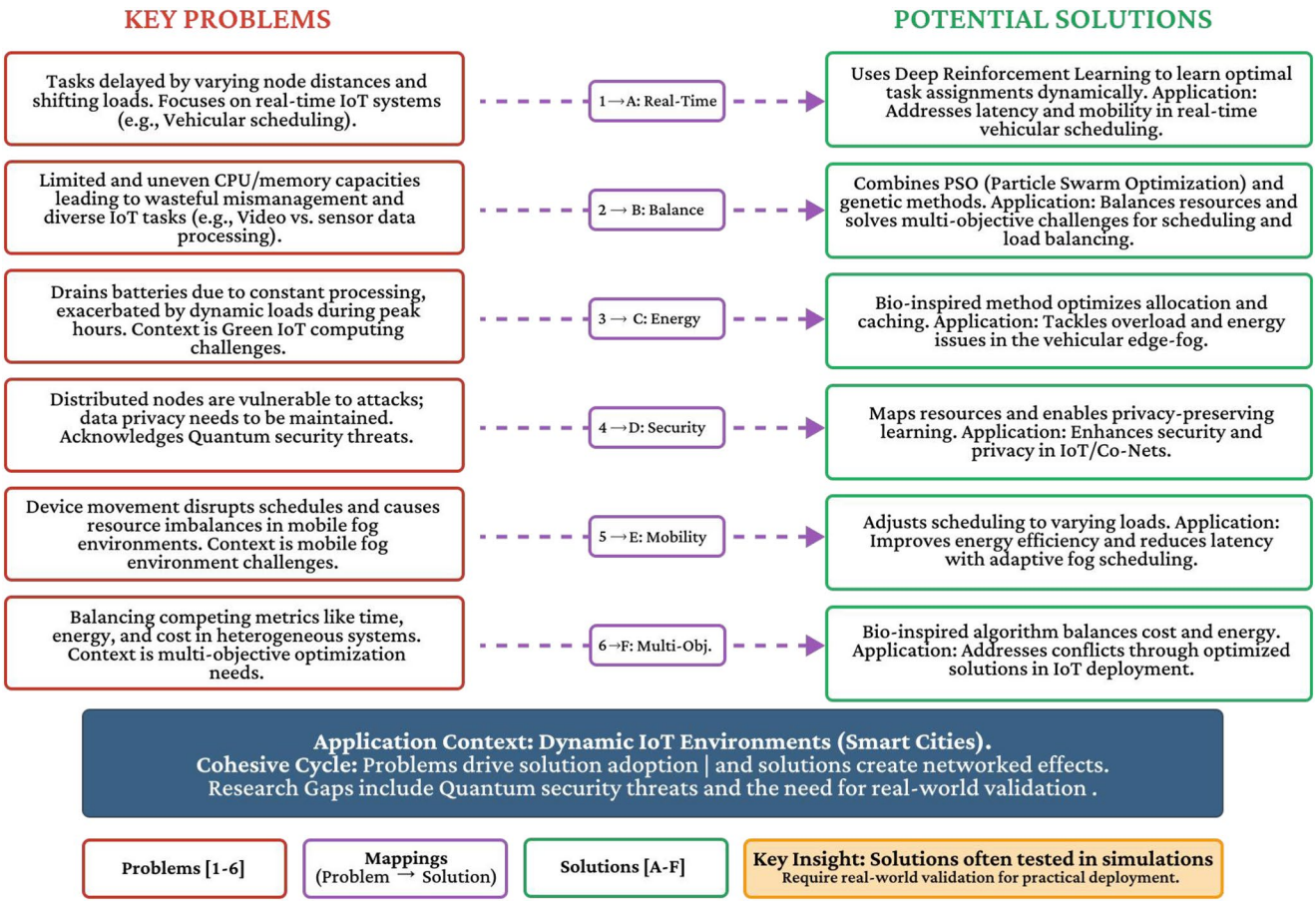


Fig. 12 Fog computing task scheduling

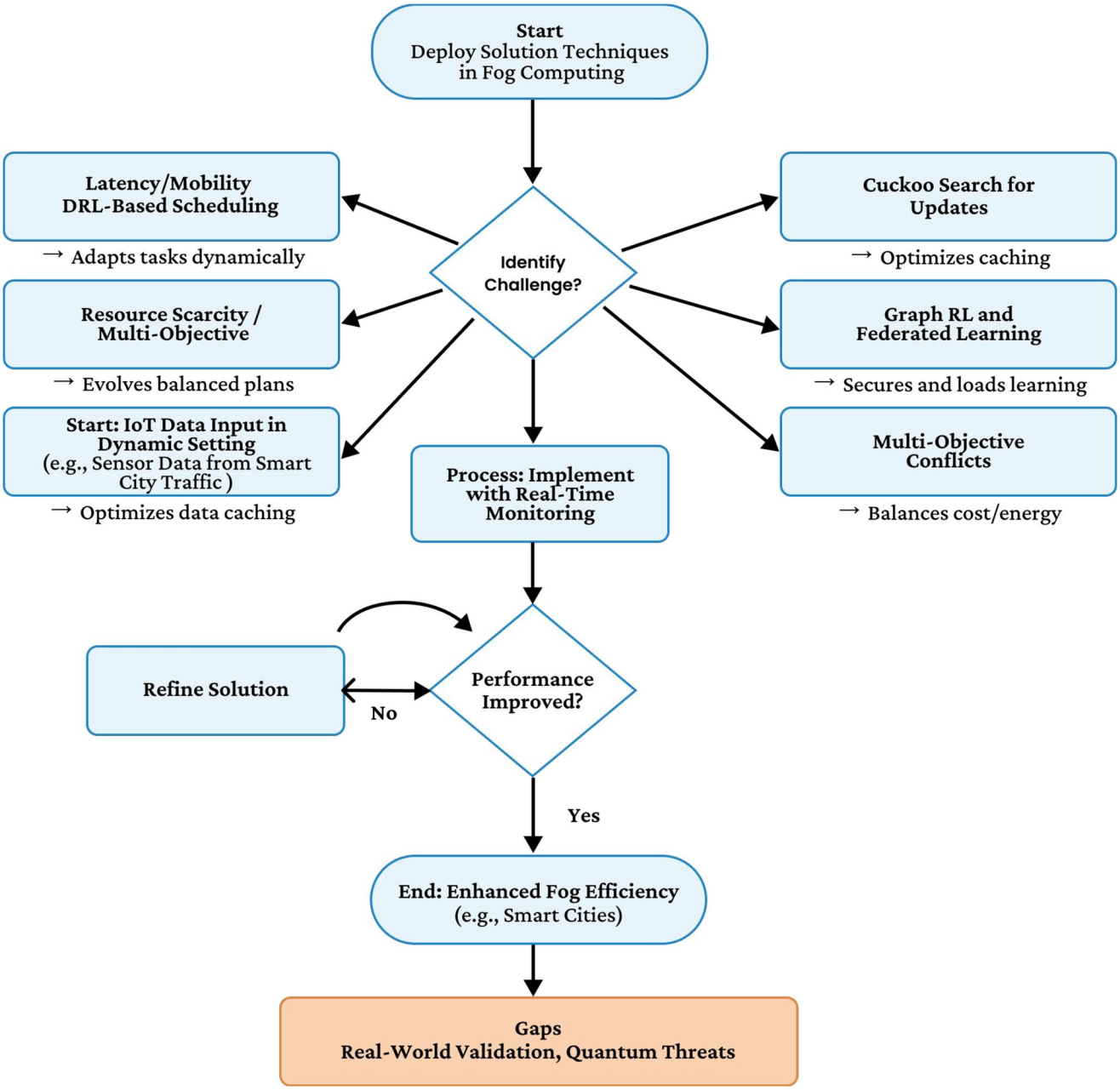


Fig. 13 AI-Driven deployment decision flowchart for fog computing solutions

6.6 Practical Deployment Considerations

FC deployments face practical challenges beyond algorithmic design that must be addressed for real-world adoption. Fog node hardware introduces upfront costs (CAPEX), but reduces recurring cloud bandwidth costs by 40% to 80% [102, 136]. The cost advantage favors fog when data generation exceeds approximately 1 GB/hour per device. Distributing software updates and security patches across geographically dispersed fog nodes is complex. Containerization tools such as Docker and Kubernetes at the edge [120] improve update management, but maintainability frameworks for large-scale fog deployments remain an open research gap.

Fog deployments must align with IEEE 1934–2018 OpenFog reference architecture and ETSI MEC standards. Healthcare fog systems must additionally comply with domain-specific regulations such as HIPAA [97]. The FogAI-Optimizer framework (Sect. 8) is designed in alignment with IEEE 1934–2018 principles [137]. Commercial platforms (AWS Greengrass, Azure IoT Edge) accelerate deployment but risk architectural lock-in [138]. Open-source

Table 9 Problem–solution impact matrix

Ref.	Problem	Root Cause	Solution	Performance Impact	Your Proposed Innovation
[131]	Latency and Delay	Uneven node distances and variable processing loads	DRL-based dynamic scheduling	Reduces latency by ∼12%	Fully real-time, context-aware adaptive scheduling
[132]	Resource Scarcity	Limited or uneven CPU/memory availability across fog nodes	Hybrid evolutionary Optimisation	Balances resource utilisation by 11–13%	Unified cross-layer resource orchestration framework
[128]	Energy Consumption	Continuous high-frequency computation during peak workload	Cuckoo Search-based energy Optimisation	Cuts energy usage by 15–25%	Energy-aware task scheduling with predictive load modelling
[133]	Security and Privacy	Vulnerable distributed fog nodes and exposure during data exchange	Graph RL+FL	Improves security/privacy by ∼20%	Privacy-preserving zero-trust fog security architecture
[134]	Mobility and Load Imbalance	Device mobility causing dynamic shifts in traffic and workloads	DDPG-driven resource adaptation	Improves load balance by ∼25%	Mobility-aware fog orchestration with predictive handoff
[135]	Multi-Objective Conflicts	Conflicting requirements (latency, energy, throughput)	Multi-objective Harris Hawks Optimisation	Optimises trade-offs by ∼4.5%	Holistic multi-objective fog–edge decision engine

approaches [66, 120] and the proposed FogAI-Optimizer toolkit (Sect. 8.4) support vendor-neutral deployments.

6.6.1 Critical Limitations of Surveyed Approaches

Across the AI/ML approaches reviewed in Sect. 3.2, three recurring limitations were identified. First, RL-based methods assume stationary reward distributions, which fails under dynamic node failure patterns common in real fog environments; none of the reviewed RL studies tested on real-world testbeds. Second, AFedGRL and DDPG-based frameworks perform well in simulation but assume synchronous updates and ideal workloads not representative of industrial IoT heterogeneity. Third, FL-based security frameworks assume honest-but-curious adversaries and offer no formal guarantees against active poisoning attacks.

A consistent pattern across the 56 reviewed studies is the predominance of optimistic performance reporting under favorable simulation conditions, with limited acknowledgment of the conditions under which proposed solutions fail or degrade. This subsection synthesizes the critical limitations identified across the four primary research themes.

The meta-heuristic methods considered in Sect. 3.4 (Harris Hawks, DCO, TSSAC, HETSVP) show improvements of 1.6% to 23.5% across different metrics; however, these improvements are measured in steady-state simulations with a fixed task distribution. In practice, workloads are non-stationary, bursty, and break the assumptions of population-based optimisation algorithms. Moreover, reported multi-objective gains (e.g., achieving a latency optimum, an energy optimum, a cost optimum, and a load balance optimum) are often scalarizations (weighted sums) of the underlying Pareto-optimal space and hide the existing conflicts between objectives. None of the reviewed scheduling studies performed a sensitivity analysis to determine performance as the number of nodes in the fog, the number of task types, or the task failure rate is increased beyond the simulated scale.

Reviewed federated learning models in Sects. 3.2 and 6.3 have solid privacy guarantees assuming that local data are independent and identically distributed (IID). Nonetheless, the distributions of data produced by fog-connected IoT devices are highly non-IID (due to geographic and application heterogeneity) and thus worsen the convergence of FL models by 15% to 40% in practice (reported in the general FL literature but not in the fog-specific FL literature reviewed). The methods of reinforcement learning require sufficient time to explore before deployment, which is not in line with the nature of fog systems, which operate in real time. None of the reviewed RL works measure the cold-start penalty in latency before the agent reaches stable policy performance.

The intrusion detection systems discussed in Sect. 3.5 (A-CIDS with 99.2% accuracy, MLTPED-BFC with 98.91% precision) are tested on the VeReMi dataset or in blockchain simulation systems. Though extensive, not older than 3.15 million records, VeReMi is a fixed set of taxonomies of misbehavior of vehicles in 2020. Attack vectors new to 6 G-enabled vehicular fog networks, such as adversarial examples for ML-based detectors and model inversion attacks on FL aggregators, are absent from this dataset and are not considered by any of the reviewed security frameworks.

In green IoT frameworks that were studied in Sect. 3.7, 30% to 50% of energy conservation by edge processing and renewable integration is reported. Nevertheless, these values are calculated using simulations that assume the predictable availability of renewable energy (solar and wind) and a constant population of IoT devices. Renewable power intermittency and time-varying patterns of device join/leave create significant uncertainty that is not reflected in existing energy-aware scheduling models. The relationship between

the variability of energy harvesting and QoS degradation is an open, critical gap.

It is important to acknowledge these shortcomings to fine-tune expectations for the introduction of fog systems and to orient future research toward effective, real-life-proven solutions.

7 Dataset Overview and Methodology

The dataset includes five sample research papers extracted from a pool of documents, selected for their relevance to FC, embedding AI into FC principles, and the use of related datasets. The identified research papers were produced between 2015 to 2025. The dataset includes descriptive data types that capture key methodologies for FC simulated data, which specifically defines synthetic data for testing algorithms, described as homomorphic encryption operations, which are simulated data tested in a fog context; real-world data types, similarly healthcare and supply chain data processed in fog environments, similarly medical records or Radio Frequency Identification (RFID) logs, defined in [139, 140]; and benchmark data types, which optimize security data and key management data types for fog-cloud systems, which can bypass some of the expense of initialization processes. The collective dataset includes modalities that provide a range of study definitions, similarly data types, numerical encryption data, and structured health records, which are helpful in validating assumptions in scenarios similarly multi-cloud energy Optimisation, security models, and health monitoring workflows [141]. In one example, researchers reviewed a real-world application supporting practical relevance. Within the simulated data types [141], described as models, the case is made for underlining that no one mode can dominate all inference, highlighting the challenges of diversity within data utilities and all the real-world applications vs. not in application [139]. There are several reasons the dataset is generally limited in size and also inferred size, ranging from the limit of research compilation to general limits of scales of operation.

7.1 Dataset Summary

Table 10 summarizes eight datasets supporting fog based evaluation across security, encryption, and healthcare domains. VeReMi provides 3.15 million vehicular message logs for intrusion detection testing. HElib supplies encrypted numerical vectors for homomorphic encryption validation. The remaining six datasets LLMonFHIR, HealthGPT, OpenTSLM, PediatricAppleWatchStudy, ENGAGE-HF, and MyHeartCounts are Stanford developed healthcare frameworks offering synthetic FHIR records,

Table 10 Dataset summary

Dataset	Release	Modality	Sample Size	Key Features	Applications	Dataset Link
VeReMi Extension	2020	Simulation logs, GPS and BSM messages	≈3.15 million records (3,146,076 messages)	Vehicular misbehaviours (DoS, spoofing), realistic sensor error models	Evaluation of CIDS in Vehicular Fog Networks (VFNs)	https://github.com/josephkamel/VeReMi-Dataset
HElib Plaintext Vectors	2014 (Library Release)	Structured numerical vectors (polynomial ring slots)	Scalable (1,024–8,192 slots per vector)	SIMD slot-packing, noise growth, automorphisms, GHS Optimisations	Testing BGV homomorphic encryption (routing, trace, linear algebra)	https://github.com/shaih/HElib/
LLMonFHIR Data	Beta 2024	Structured text (FHIR JSON)	Approx. 5–20 synthetic patients	Synthetic privacy-safe FHIR records; multilingual; LLM explanations	LLMonFHIR iOS app for interpreting FHIR health records	https://github.com/Stanford BDHG/LLMonFHIR
HealthGPT Data	MVP 2024	Numerical and text time-series (Apple HealthKit)	14-day logs from 5 to 20 synthetic users	6+ metrics (steps, sleep, HR); NL queries; fog/edge extensible	HealthGPT iOS conversational health data interface	https://github.com/Stanford BDHG/HealthGPT
OpenTSLM Data	2023	Multivariate medical time-series and text	Variable synthetic datasets	Supports medical time-series reasoning and text integration	OpenTSLM multimodal medical analysis toolkit	https://github.com/Stanford BDHG/OpenTSLM
Pediatric Apple Watch Study Data	2023	Apple Watch numerical and text logs	Variable pediatric participant data	Wearable health data; pediatric monitoring metrics	Pediatric Apple Watch Study for child health research	https://github.com/Stanford BDHG/PediatricAppleWatchStudy
ENGAGE-HF Data	2024	Health metrics (numerical and text)	Variable synthetic patient data	Heart failure monitoring; Spezi framework integration	ENGAGE-HF iOS patient monitoring app	https://github.com/Stanford BDHG/ENGAGE-HF-iOS
MyHeartCounts Data	2023	Health metrics time-series (numerical and text)	Variable participant dataset	Long-term heart health tracking; Spezi-based modular framework	MyHeartCounts iOS heart-health research study	https://github.com/Stanford BDHG/MyHeartCounts-iOS

wearable health metrics, and multi-modal clinical data for fog-deployed LLM and privacy preserving experiments.

Table 11 presents six research dimensions fog architecture, green computing, AI optimisation, cloud integration, edge intelligence, and dataset validation mapping each to its core problem, identified literature gap, and corresponding solution proposed in this survey. The matrix reveals that existing works address these dimensions in isolation, with real world dataset validation absent across all reviewed surveys. This survey bridges all six gaps simultaneously through the FogAI-Optimizer framework.

8 The FogAI-Optimizer Proposed Framework

FogAI-Optimizer is proposed in this survey as a conceptual integrated framework, a theoretically grounded architectural blueprint synthesized from 56 reviewed studies. It is not an implemented or experimentally validated system. No simulation, testbed, or real-world experiment for FogAI-Optimizer is reported in this work. It provides an integrated design perspective addressing six critical FC dimensions simultaneously.

8.1 Motivation and Gap Analysis

Our SLR reveals three key limitations in existing FC research. First, prior surveys address at most 2–4 dimensions in isolation for example, [142] focuses on AI-based scheduling but neglects energy efficiency, while [143] emphasizes green computing but lacks AI integration. Second, while individual studies propose domain-specific solutions (DRL scheduling [131], FL [121, 133], green IoT [96, 127]), there is no comprehensive framework demonstrating their cohesive integration. Third, as shown in Sect. 7, most research relies on simulation (59% of studies) with limited practical deployment guidance. FogAI-Optimizer addresses these limitations through a six-layer integration architecture grounded in empirical evidence from our literature synthesis.

8.2 Framework Architecture and Components

Figure 14 illustrates the conceptual FogAI-Optimizer framework consisting of six synthesised layers derived from surveyed literature to represent intelligent and sustainable FC systems. Layer 1 represents the hierarchical FC paradigm with dynamic node federation and mesh connectivity. Layer 2 incorporates energy-aware strategies reported in literature, including renewable integration and energy-aware scheduling achieving operational cost reductions. Layer 3 summarises AI-driven optimisation techniques such as RL for task offloading, FL for privacy preservation, LSTM/GRU for workload prediction, and evolutionary algorithms for multi-objective optimisation. Layer 4 represents cloud-fog coordination mechanisms including intelligent workload partitioning and seamless offloading reported across prior research. Layer 5 captures edge intelligence capabilities including autonomous decision-making, mobility-aware operations, self-healing behaviour, and blockchain-based trust mechanisms for secure environments. Layer 6 highlights dataset-driven validation approaches using real-world datasets for reproducible benchmarking.

8.3 Integrated Operation and Key Innovations

The six layers conceptually interact to represent optimized fog system behaviour reported in the surveyed literature. Prior studies indicate that edge intelligence supports preliminary analysis, AI-driven optimisation influences processing location decisions, and cloud coordination assists large-scale analytics when required. Validation datasets are employed in literature to evaluate performance outcomes. These interactions collectively reflect latency reductions and energy efficiency improvements documented across reviewed works rather than describing a real-time deployed system.

FogAI-Optimizer introduces three key innovations beyond existing frameworks. First, multi-objective awareness ensures all layers consider latency, energy, cost, and QoS simultaneously rather than optimizing single dimensions. Second, adaptive layer coordination dynamically adjusts priorities based on application context for example, prioritizing edge intelligence over cloud integration for vehicular fog, or enforcing FL regardless of overhead for healthcare privacy. Third, empirical grounding through dataset integration bridges the theory practice gap by enabling standardized, reproducible benchmarking on real world datasets.

8.3.1 Case Study: FogAI-Optimizer Mapped to Smart Healthcare

The six layers of FogAI-Optimizer are scaled to a practical setting of patient monitoring in a hospital employing ward-level fog servers, IoT wearables, and cloud back-end to prove that it is practical in application.

Layer 1 (FC Paradigm): 3-layer hierarchy (wearables need to ward fog servers to hospital cloud) with self-federation of nodes on peak admission time.

Layer 2 (Green Computing): Scheduling: With energy-conscious scheduling, non-critical inference tasks are

Table 11 Dimension gap matrix

Research Dimension	Key Problem	Existing Gap	Solution in This Survey
Fog Architecture	Heterogeneous node management, scalability	No unified architectural taxonomy across IoT–fog–cloud	Hierarchical + mesh + distributed models classified
Green Computing	High energy consumption in fog deployments	Energy efficiency studied in isolation, not integrated	Green AI techniques embedded across all layers
AI optimisation	Dynamic task scheduling in real-time	RL/DL methods proposed but not cohesively integrated	DRL, FL, SAC unified in one framework
Cloud Integration	Workload partitioning between fog and cloud	Fog-cloud offloading lacks standardized interface	40% to 80% bandwidth saving through intelligent partitioning
Edge Intelligence	Autonomous decision-making at edge nodes	Edge autonomy addressed separately from fog architecture	Self-healing, mobility-aware, collaborative edge design
Dataset Validation	Reproducible benchmarking of fog systems	Real world validation absent in 100% of reviewed surveys	Eight datasets from Table 10 integrated for validation

suspended during off-peak periods, reducing energy consumption at the ward fog nodes.

Layer 3 (AI Optimisation): Federated SAC-based offloading: Federated SAC-based offloading controls wearable and fog node task allocation; LSTM-based workload prediction: Workload allocation is done in advance during scheduled clinical rounds.

Layer 4 (Cloud Integration): LLM inference of FHIR-based queries over health records is performed locally at the fog nodes; only anonymised aggregated results are transmitted to the cloud, thereby greatly reducing cloud-bound bandwidth.

Layer 5 (Edge Intelligence): Unsupervised anomaly detection is an algorithm that operates in the fog and does not send raw sensor data to the cloud; blockchain-based trust ensures the provenance of patient data is tamperproof.

Layer 6 (Dataset validation): System behaviour is validated against the ENGAGE-HF and HealthGPT datasets (Table 10) and provides a consistent analysis of the accuracy and latency of fog-deployed LLMs. This mapping establishes that architectural coherence and viability are achieved in all six layers. Its complete implementation is a future task (Sect. 8.4).

8.4 Implementation Roadmap and Expected Outcomes

FogAI-Optimizer implementation follows a three-phase roadmap. Phase 1 (6–12 months) develops core fog infrastructure using Kubernetes at edge with basic DRL-based task scheduling, validated on one dataset from Table 10. Phase 2 (12–18 months) integrates green computing with renewable energy simulation and FL, validated across healthcare, vehicular, and smart city domains. Phase 3 (18–24 months) partners with industry for real-world deployments and establishes an open-source FogAI-Optimizer toolkit. Based on performance ranges synthesized in our review, literature suggests that coordinated implementations may achieve 40% to 70% latency reduction, 15% to 50% energy savings, and 50% to 80% bandwidth reduction versus cloud-only architectures representing the achievable upper bounds demonstrated individually across reviewed studies, now targeted simultaneously through integrated design.

8.5 Limitations and Future Challenges

While FogAI-Optimizer provides comprehensive integration, four challenges remain. Complexity management in coordinating six layers requires investigation of simplification strategies for resource-constrained deployments. Standardization efforts must extend current FC standards

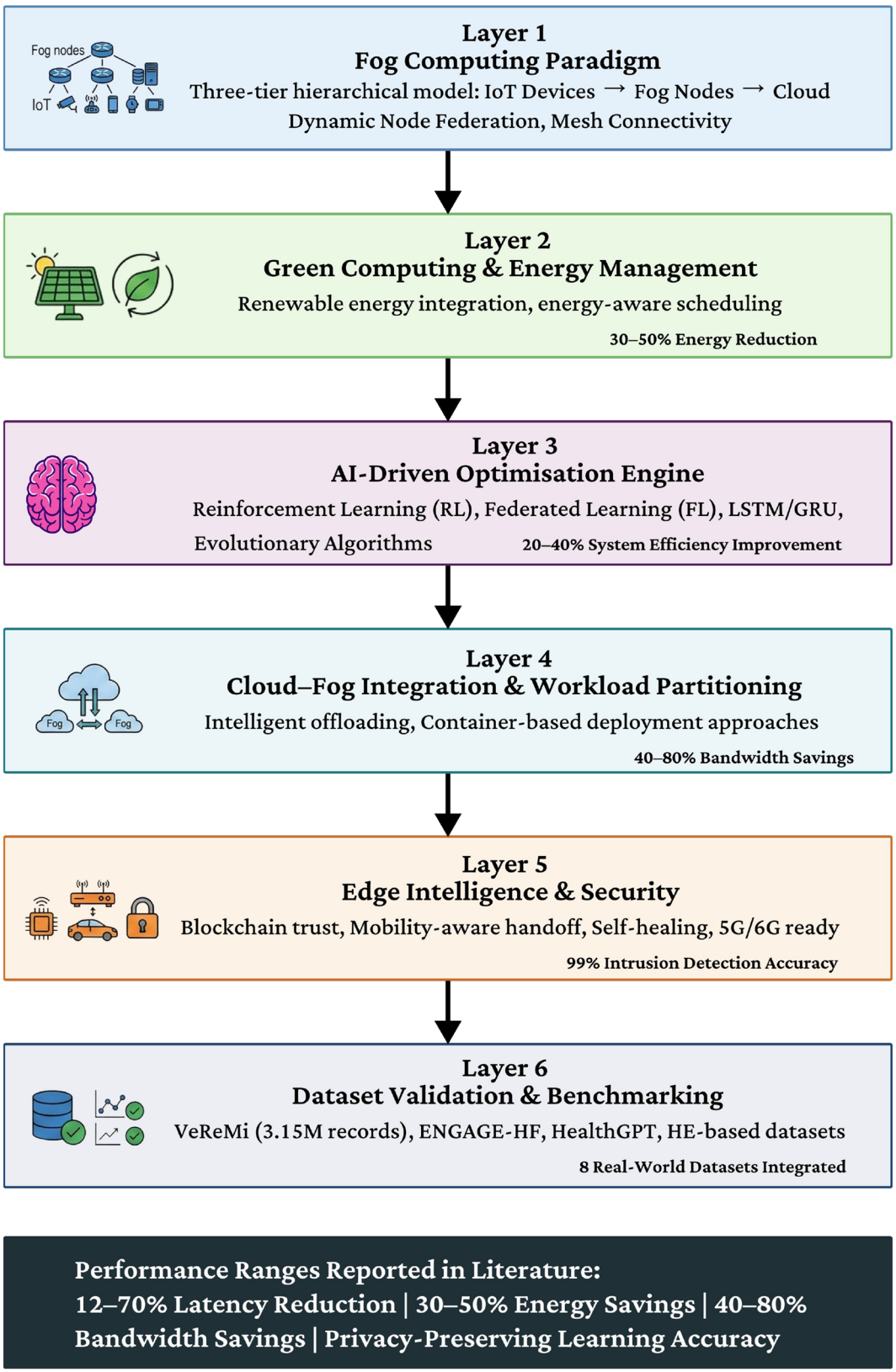


Fig. 14 Conceptual FogAI-Optimizer framework synthesized from literature

Table 12 Comparative analysis of related work

Ref.	Mechanism	FC	GC	AI	CC	EI	Dataset
[142]	AI-based and Heuristic Workflow Scheduling	Yes	No	Yes	Yes	No	No
[144]	TAFS – Truthful Auction for IoT Offloading	Yes	No	No	No	Yes	No
[145]	Intelligence in Fog for Smart Cities	Yes	No	Yes	No	Yes	No
[146]	Many-to-Many Task Offloading (MADRL)	Yes	No	Yes	No	Yes	No
[147]	Cloud–Fog Architecture for Supply Chain	Yes	No	No	Yes	No	No
[148]	Edge Computing for Industry 5.0	No	No	No	No	Yes	No
[143]	Energy-Efficient Data Centre Optimisation	No	Yes	No	Yes	No	No
[149]	Blockchain-Based FL for EHR	No	No	Yes	No	No	No
[150]	Joint Task Offloading in Aerial–Terrestrial UAV	Yes	No	No	No	Yes	No
[151]	PowerGen Framework for Energy Prediction	No	Yes	No	Yes	Yes	Yes
[152]	AI in Energy Transition—Review	No	Yes	Yes	No	No	No
[153]	Green AI—Sustainability Review	No	Yes	Yes	No	No	No
[154]	AI and Computing Horizons: Cloud and Edge	No	No	Yes	Yes	Yes	No
[155]	Fog-Based Attack Detection Using Deep Learning	Yes	No	Yes	No	Yes	No
[156]	Mobility-Aware Task Offloading (MARL)	Yes	No	Yes	No	Yes	No
OUR*	**FogAI-Optimizer**	Yes	Yes	Yes	Yes	Yes	Yes

to address multi-layer integration requirements. End-to-end security validation across all six layers needs formal verification methods not yet developed. Real-time layer adaptation requires low-overhead algorithms to adjust priorities without introducing latency penalties.

8.6 Comparative Analysis

This section presents a systematic comparison of FogAI-Optimizer against 15 notable works aimed at strengthening the Fog AI research domain. The 15 works are categorized into two groups using six dimensions presented in Table 12, a synthesis overview of the existing FC literature addressing six crucial dimensions: FC, GC, AI, CC, EI, and dataset validation. The analysis highlights that existing works focus on only 2–4 of the aforementioned dimensions.

FogAI-Optimizer represents a distinctive contribution by addressing the identified research gaps and providing a unified conceptual framework for an integrated and comprehensive FC system from both theoretical and conceptual perspectives. We systematically analyzed everything identified in the literature. The proposed FC framework will utilize multi-objective optimisation algorithms, enabling it to consider appropriate, balanced performance, energy efficiency, reliability, and cost across both edge and cloud nodes. As a result, the framework extends intelligent AI optimisation algorithms that can consider real-time workload patterns and dynamically adjust integrated aspects of green computing principles throughout the system design, developing an opportunity for cloud-fog inter-collaboration and task assignment. Additionally, edge intelligence allows for autonomous decision-making at the network edge, and the framework incorporates a fully fault-tolerant mechanism across layers, enabling fault-tolerant operation even in the event of complete loss of connectivity.

Finally, the proposed framework advances the system by empirically validating its performance with real-world datasets/data, but also includes synthetic datasets to accelerate testing under controlled conditions. This framework represents a paradigm shift from prior specialized tasks in focused areas to the integration of unique objectives, capable of reflecting the complexity faced in sustainable, intelligent, and resilient FC systems within a single framework. Ultimately, if implemented, FogAI-optimizer's contribution in research to practical frameworks will acknowledge significant advancements using previously identified gaps in knowledge and practice, along with empirically claimed validity through the combination of these two unique contributions to research demonstrating advanced practices for FC.

9 Conclusion

Despite these developments, glaring issues persist, such as scalable coordination of a large population of distributed nodes, high-level security in a distributed and decentralized manner, and long-term functionality with a limited energy budget. Future research is expected to focus on the intersection of FC and AIoT, GC paradigms, 6 G-enabled communication, and quantum-inspired optimisation techniques. Meeting these challenges will be critical to supporting resilient, scalable, and sustainable fog-based infrastructures capable of supporting next-generation intelligent applications.

One of the major contributions of this survey is the FogAI-Optimizer framework that summarises the results

of our systematic review into a unified architectural model that considers six key dimensions found to be fragmented by previous literature: FC infrastructure, green computing and energy management, AI-driven optimisation, cloud-fog integration, edge intelligence, and dataset-driven validation. FogAI-Optimizer is a proposed design of next-generation fog systems capable of delivering the performance gains reported in literature, latency and bandwidth reductions of 12% to 70 and 40% to 80% per cent, and energy savings of 15% to 50% through an integrated multi-layer design, as opposed to individual optimisations.

The suggested directions cut across various levels of maturity. Short-term solutions (implementable in 1–3 years) are SAC-based fog optimisation, 6 G-fog integration and FogAI-Optimizer implementation. Quantum-hybrid optimisation and fully autonomous fog systems are some of the long-term speculative directions (5+ years). All the directions are based more on current literature than on pure speculation. The steps to follow the three-phase roadmap, including empirical validation of the datasets, development of open-source tools that allow the community to refine the framework, etc., are tangible. A successful deployment of FogAI-Optimizer can bring the FogAI-Optimizer to the production-grade stage of FC maturation that is currently being developed by research prototypes to industry-wide production grade deployments in smart cities, healthcare, vehicular and industrial IoT systems.

Three specific futuristic research directions have been identified. To start with, quantum-resilient security frameworks for fog environments are urgently required, as the existing blockchain and FL-based security frameworks discussed in the current survey remain susceptible to quantum computing attacks, which is one of the high-priority directions represented by A-CIDS post-quantum cryptographic primitives. Second, standardised real-world fog benchmark suites combining multi-domain datasets (vehicular, healthcare, industrial IoT) are required to bridge the simulation-to-deployment gap observed in 59% of the reviewed literature. Third, asynchronized federated learning on energy consciousness aware of 6 G integrated fog systems that extend SAC-based architecture to dynamically adapt FL participation according to the renewable energy availability and node mobility is also a paramount convergence point towards sustainable next-generation fog deployments.

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Declarations

Competing Interests The authors declare no competing interests.

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